



Statistics Department
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"Farm-Level Determinants of Wheat Productivity, Post-Harvest Losses, and Profitability in Egypt"

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"وَأَن نُّنِيسَ لِلنَّفْسَانِ إِلَٰهَ مَا سَعَىٰ"

النجم (٣٩)

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Abstract

Wheat is a strategic crop in Egypt due to its importance for food security and household consumption, yet domestic production remains insufficient to meet national demand. Therefore, understanding farm-level performance is important for improving production efficiency and reducing losses.

This study uses secondary data from the FAO Good Growth Plan survey for Egypt during the 2022–2023 wheat season. The dataset covers production practices, input use, costs, water source, crop damage, post-harvest conditions, profitability, and seed retention decisions. Quantile regression, logistic regression, and Factor Analysis of Mixed Data are applied to examine the relationships between farm characteristics and production outcomes.

The quantile regression results show that factors associated with wheat yield differ across productivity levels. Pesticide cost per hectare is positively and significantly associated with yield across the selected quantiles, while water source, seed cost, damaged percentage, spray guidance, and total fallow land vary in importance across farmer groups.

The logistic regression results indicate that rain-fed farming and seasonal accidents are associated with a higher likelihood of high post-harvest loss, while market-based input sources are associated with lower loss risk. Profitability is shaped by production conditions, cost structure, pest exposure, water source, and seasonal shocks, with seasonal accidents showing the clearest negative association. Seed retention decisions are negatively associated with yield per hectare and seed cost per hectare.

Finally, Factor Analysis of Mixed Data identifies three main dimensions of farmer differences: technical scale and input intensity, agronomic innovation and resource strategy, and resource optimization with economic perception.

Overall, the study shows that Egyptian wheat farmers differ substantially in their production conditions, management practices, and economic outcomes, highlighting the importance of considering farm-level variation when evaluating wheat performance.

Keywords: Wheat, Quantile regression, logistic regression, Factor Analysis of Mixed Data.

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List of Acronyms

Acronym	Full Name
AIC	Akaike Information Criterion
ARDL	Autoregressive Distributed Lag
AUC	Area Under the Curve
CAPI	Computer-Assisted Personal Interviews
FAM	Food and Agriculture Microdata
FAMD	Factor Analysis of Mixed Data
FAO	Food and Agriculture Organization of the United Nations
GGP	Good Growth Plan
GLM	Generalized Linear Model
GVIF	Generalized Variance Inflation Factor
IFPRI	International Food Policy Research Institute
LR	Logistic Regression
MCA	Multiple Correspondence Analysis
MLE	Maximum Likelihood Estimation
OLS	Ordinary Least Squares
PCA	Principal Component Analysis
QR	Quantile Regression
ROC	Receiver Operating Characteristic
VIF	Variance Inflation Factor.

Chapter (1)

Introduction

1.1 Overview:

Wheat is one of the most important staple crops worldwide and represents the backbone of food security, particularly in developing and food-importing countries. According to the Food and Agriculture Organization (FAO), wheat provides about 20 percent of the total dietary calories consumed globally, making it a critical component of human nutrition and economic stability. In Egypt, wheat holds exceptional strategic importance, as it constitutes the main ingredient in subsidized bread, a key element of the national food security system, and a primary source of calories for a large share of the population.

Despite its importance, domestic wheat production in Egypt remains insufficient to meet rapidly growing demand driven by population growth, urbanization, and changing consumption patterns. As a result, Egypt has become one of the world's largest wheat importers, increasing its vulnerability to international price volatility, supply chain disruptions, and external shocks. The World Bank (2023) emphasizes that heavy reliance on wheat imports exposes Egypt to significant fiscal and food security risks, particularly during periods of global instability, such as climate shocks, pandemics, and geopolitical conflicts. Consequently, improving domestic wheat productivity and reducing production and post-harvest losses have become national priorities (World Bank, 2023).

At the farm level, wheat production outcomes in Egypt are characterized by substantial variability. Farmers differ widely in yields, cost structures, post-harvest performance, and profitability. These differences are influenced by a complex interaction of agronomic practices, input use decisions, access to fertilizers and crop protection products, irrigation sources, exposure to adverse weather conditions, and the adoption of safety and post-harvest management practices. The FAO (2019) reports that inefficiencies in input use and inadequate post-harvest handling contribute significantly to food losses in cereal crops across developing countries, reducing both farmer incomes and national food availability (FAO, 2019).

Post-harvest losses represent a particularly critical challenge in wheat production. Studies estimate that a substantial share of wheat losses in Egypt occurs during harvesting, storage, and handling, leading to wasted resources such as water, energy, and labor, while also increasing the country's import needs (Yigezu et al., 2021). Reducing these losses is often more cost-effective than expanding cultivated areas,

as it directly enhances food availability without placing additional pressure on scarce land and water resources. Moreover, improved post-harvest practices can significantly increase farm-level profitability and income stability.

In addition to productivity and losses, farmer decision-making plays a crucial role in shaping production outcomes. Decisions related to fertilizer application, crop protection, soil testing, seed saving, adoption of improved seed varieties, and compliance with pesticide safety guidelines affect not only yields and profits but also long-term sustainability and farmer health. The International Food Policy Research Institute (IFPRI, 2023) highlights that access to agronomic advice and reliable input markets are essential for encouraging the adoption of improved technologies and practices among smallholder farmers in developing countries.

This study uses FAO's Good Growth Plan microdata for Egypt (2022–2023) to identify the key farm-level determinants of wheat productivity, post-harvest losses, profitability, and farmer decision-making, by providing empirical evidence from nationally collected microdata.

1.2 Literature Review:

- **Agronomic, Climatic, and Input-Related Drivers of Wheat Yield**

Alboghady et al. (2025) examined the short- and long-term impacts of climate change and agricultural inputs on wheat production in Egypt (1961–2022) using an Autoregressive Distributed Lag (ARDL) model and Granger causality analysis. The study showed that expansion of harvested area, technological improvements, and moderate winter temperature increases boosted long-term wheat production, while higher CO₂ emissions and adverse spring conditions reduced short-term productivity. The analysis also confirmed a stable long-term equilibrium and adaptive capacity in Egypt's wheat sector.

El Wakeel and Yassa (2025) examined how different wheat cultivation methods affected total factor productivity and economic performance in Beheira Governorate, Egypt. Their study found that terraced cultivation increased total productivity of production factors compared with traditional methods, reducing variable costs by about 4.6% and enhancing profitability per feddan. Terraced farming ¹also reduced seed

¹ Terraced cultivation (also known as bed planting or ridge cultivation) is a non-traditional wheat farming method in which seeds are planted on raised beds or furrows, as opposed to conventional flat-surface broadcasting. This technique improves water distribution, reduces seed waste, and facilitates mechanized management (El Wakeel & Yassa, 2025).

requirements by 16.2%, saved roughly 349 m³ of irrigation water per feddan, and increased wheat productivity by about 15.6%, demonstrating that improved cultivation techniques could enhance both resource efficiency and economic outcomes for wheat producers.

Vishwakarma et al. (2025) examined the drivers of global wheat yield shocks using a panel dataset covering 86 countries over a 30-year period from 1979 to 2014. The study employed two complementary quantile regression approaches which are a linear quantile mixed model and a quantile random forest to assess the relationships between yield shocks and a set of ecological and socioeconomic drivers including extreme weather stress, nitrogen fertilizer application, GDP per capita, agricultural GDP, and irrigated area. The results revealed that the severity of wheat yield shocks has been increasing globally since 1997, and that extreme weather stress was the most consistently important driver across all quantile levels. Notably, countries heavily reliant on irrigation including Egypt, where up to 100% of cropland is irrigated showed a levelling off in shock severity, highlighting the complex role of water management in agricultural resilience.

Abdalla et al. (2023) examined how agronomic practices affected wheat productivity in Egypt's Nile Delta. Key yield determinants included irrigation frequency, crop rotation, and fertilizer use, while land size and labor also positively influenced output. Excess fertilizer use showed little yield benefit, highlighting input inefficiencies. The study emphasized that balanced fertilization, improved water management, and access to agronomic knowledge were essential to sustainably increase wheat yields.

Hussain et al. (2023) investigated the impact of climate volatility on wheat productivity and food security in the Faisalabad district of Pakistan using daily temperature and rainfall data spanning 33 years. The study applied quantile regression to estimate the influence of climate volatility across the full distribution of wheat yields, rather than focusing solely on average effects. The results provided evidence that climate volatility was more damaging to wheat production than gradual changes in temperature or rainfall, with effects varying considerably across yield quantiles. The findings highlighted the advantage of quantile regression in capturing asymmetric impacts across low- and high-performing farms, making it a valuable analytical tool for studying yield determinants in contexts characterized by wide productivity variability and skewed distributions, as is the case in Egyptian wheat production.

- In alignment with these methodologies, our study also applied quantile regression to examine how various agronomic, environmental, and technical input variables influence Egyptian wheat yield per hectare across the 10th, 50th, and 90th quantiles. The results confirm that the determinants of

wheat yield are highly heterogeneous and vary considerably based on the farm's productivity segment. Specifically, pesticide investment per hectare is the only universally significant driver that positively influences productivity across all three selected quantiles.

- **Determinants and Stages of Post-Harvest Wheat Losses**

Nath et al. (2024) reviewed global technologies and practices for reducing postharvest grain losses, highlighting that poor handling, drying, and storage led to both quantity and quality losses. Modern solutions such as mechanical drying, hermetic storage, and improved handling effectively reduced losses, but adoption was limited by cost, training gaps, and lack of extension support. The study emphasized that sustainable postharvest loss reduction required both technological innovation and capacity building, offering valuable insights for countries like Egypt aiming to improve wheat storage and reduce losses.

El-Batran et al. (2023) studied wheat post-harvest losses in Ismailia Governorate, Egypt, covering 368 farmers during the 2021/2022 season. Total losses across the sample reached approximately 318 tons, distributed across five post-harvest stages: harvesting (106 tons, 33.4%), drying (84 tons, 26.5%), threshing (58 tons, 18.2%), packaging (41 tons, 12.8%), and transportation (29 tons, 9.1%), equivalent to about 2.08% of total production and an average loss of 403 kg per feddan. The study recommended early harvesting, wider adoption of combine harvesters, and strengthened agricultural extension services to reduce losses and improve farm income.

Yigezu et al. (2021) conducted a comprehensive value chain and resource footprint analysis² of food losses along Egypt's wheat value chain, finding that about 20.6% of the total supply (~4.4 million tons) was lost in 2017/2018. Losses occurred at all stages, wasting food, water (~4.79 billion m³), and energy (~74.72 million GJ). The study emphasized that reducing post-harvest losses could enhance food availability, save foreign exchange, lower imports, and reduce environmental impacts, highlighting the need for improved handling, storage, and policy interventions.

Huang et al. (2017) applied logistic regression to analyze the determinants of grain harvest step losses in China, using data from 5,400 samples across 31 provinces over the period 2012 to 2014. The study examined six key factors influencing harvest losses weather conditions, harvest timing, pest maturity, field

² Value Chain Analysis: A methodology that tracks the flow of a commodity through all stages of production and distribution to identify points of loss and value addition.

Resource Footprint: A measure of the total environmental impact in terms of natural resources (specifically water and energy) required to produce a specific quantity of a good. (Yigezu et al., 2021).

transportation, threshing method, and grain cleaning method and used a stochastic gradient descent algorithm to optimize model parameters. The logistic regression model achieved a prediction accuracy of 86.25%, which improved to 92.53% following optimization. The findings confirmed that harvest timing, transportation mode, and threshing method were among the most influential factors in determining the extent of grain harvest losses, underscoring the importance of post-harvest management practices in reducing food losses at the farm level.

- Mirroring this analytical framework, our study utilized binary logistic regression to evaluate the farm management, biological, and environmental factors driving high post-harvest wheat loss levels in Egypt. Our findings confirm that environmental exposure and production infrastructure are primary forces in shaping post-harvest vulnerability. Water management stands out as the most powerful statistically significant predictor.
- **Production Costs, Economic Efficiency, and Profitability in Wheat Farming**

Shaheen et al. (2025) analyzed the economic and marketing efficiency of wheat production in Egypt. They highlight that although wheat was widely cultivated, domestic production only achieved about 46% self-sufficiency in 2023, forcing heavy reliance on imports of about 9.212 million tons in 2024. The study showed that expanding the cultivated area by 500,000 feddans could increase production and reduce the consumption gap, potentially raising self-sufficiency to over 55%. These findings suggested that increasing cultivated area and improving marketing systems were necessary to enhance Egypt's wheat supply, reduce dependency on imports, and strengthen the overall economic performance of the wheat sector.

Bucheli et al. (2024) examined how precipitation near harvest affected wheat quality and economic losses, focusing on downgrading from low Hagberg Falling Number³. They used Swiss winter wheat trial data (1,859 observations, 2008–2019) and a two-way fixed effects model. The study found that rainfall in the 31 days before harvest increased the probability of quality downgrading by about 0.1 percentage points per millimeter. This downgrading could reduce farm income by up to CHF 1,445 per hectare, highlighting the economic impact of weather-induced quality risks in wheat production.

³ The Hagberg Falling Number (HFN) is an internationally standardized industry measure of wheat baking quality. It measures the time (in seconds) for a stirrer to fall through a mixture of flour and water. Higher values indicate better baking quality. In Switzerland, wheat scoring below 220 seconds is downgraded from bread/biscuit wheat to the lower-priced animal feed category, triggering significant revenue losses for farmers (Bucheli et al.,2024).

El Bahgy et al. (2024) examined the economic impact of adopting improved wheat technologies among Egyptian smallholder farmers using data from 1,266 households across ten governorates. The study found that adopting improved wheat varieties increased yield by about 359 kg per feddan and boosted farmer income by around 1,770 Egyptian pounds per feddan compared to non-adopters. Adoption also raised individual wheat consumption by approximately 23%, indicating benefits for both productivity and food security. The authors recommended promoting access to improved varieties and strengthening extension services to enhance adoption and increase national wheat productivity.

- **Seed Source Decisions and Farm-Saved Seed Behavior Among Wheat Farmers**

Byamungu et al. (2026) examined what determines whether wheat farmers choose farm-saved or certified seed. Using logit, latent class, and propensity score matching models on 450 wheat farmers, it found that economies of scale and knowledge-based factors, including awareness, access to extension services, and education, consistently shape seed choice decisions, with significant heterogeneity across socio-economic classes.

Minwagaw (2021) analyzed what drives smallholder wheat farmers to use farm-saved seed versus formal certified seed systems. Using a binary logit model, it found that seed quality, educational level, extension contact, access to credit, and household income were statistically significant positive factors, while distance from seed distribution areas negatively influenced the probability of selecting the formal seed system. It was also found that 100% of producers confirmed that sources of bread wheat seed were from farm-saved and local market sources which were uncertified, leading to production deterioration.

- Following a matching empirical strategy, our research employed a binary logistic regression model to explore the core economic and productivity drivers underpinning Egyptian wheat farmers' choices to engage in seed retention behavior. The estimation reveals that seed-saving decisions are governed by deliberate economic and productivity-driven trade-offs rather than mechanization costs or unpredictable farm shocks.

1.3 Study problem:

Wheat is a strategic staple in Egypt, critical for food security and rural livelihoods, yet domestic production fails to meet growing demand. As a result, Egypt has become one of the world's largest wheat importers, heavily dependent on foreign supplies. Farmers face wide variability in yields, post-harvest losses, and profitability, driven by complex interactions between agronomic practices, input access, safety measures, and environmental shocks. Despite its importance, these factors remain poorly understood. This study seeks to identify the key determinants of wheat productivity and post-harvest performance, providing evidence to enhance domestic production, reduce losses, and strengthen farmer livelihoods.

While existing studies have shed light on selected dimensions of Egypt's wheat sector, critical gaps remain. Research has examined the role of climate change and agricultural inputs on long-run wheat productivity (Alboghdady et al., 2025), the economic benefits of improved cultivation techniques (El Wakeel and Yassa, 2025), and the scale of post-harvest losses along the value chain (Yigezu et al., 2021; El-Batran et al., 2023). Others have highlighted the persistent self-sufficiency deficit and the potential of area expansion to reduce import dependency (Shaheen et al., 2025), as well as the productivity gains associated with improved variety adoption (El Bahgy et al., 2024). However, no study has simultaneously examined yield determinants, post-harvest loss drivers, production cost structure, seed-saving behavior, and the latent dimensions that differentiate farmer profiles within a single integrated framework. This gap limits the ability of policymakers and extension services to design targeted, evidence-based interventions that address the full complexity of Egypt's wheat farming system.

1.4 Objectives:

General Objective

The general objective of this study is to examine the determinants of wheat yield, post-harvest losses, profitability, and farmer decision-making in Egypt using farm-level survey data.

Specific Objectives

- 1. Yield Determinants:** To identify the agronomic, environmental, and input-related factors (such as fertilizer use, crop protection, irrigation source, adverse weather, soil testing, and agro-advice) that influence wheat yield among Egyptian farmers.

2. **Post-Harvest Loss Determinants:** To examine the key biological, environmental, and farm management factors driving high post-harvest wheat loss among farmers and evaluate how these factors affect the probability of experiencing high loss levels.
3. **Production Costs and Profitability:** To analyze how production costs and cost structure influence farm profitability and identify which cost components significantly affect the likelihood of profitable operations.
4. **Seed-Saving Decisions:** To investigate how farm and storage factors, production outcomes, and post-harvest practices affect farmers' decisions to save seed or retain a portion of their harvest.
5. **Latent Structure of Farming Systems:** To uncover the underlying latent dimensions that jointly capture the interplay between agronomic practices, economic outcomes, post-harvest behavior, and farmer perceptions among Egyptian wheat farmers.

Chapter (2)

Data and Methodology

2.1 Data Framework and Structure:

This study relies on secondary data obtained from the FAO Good Growth Plan (GGP) survey for Egypt, conducted during the 2022–2023 wheat season (FAO, 2023).

2.1.1 Data Overview:

This study employs a cross-sectional observational design based on secondary data collected through structured surveys of wheat farmers in Egypt. The data were obtained from the post-harvest wheat micro dataset available through the FAO Food and Agriculture Microdata (FAM) Catalogue, collected as part of the FAO Good Growth Plan survey for the 2022–2023 wheat production season. The dataset originally consists of 398 farm-level observations and 57 variables covering post-harvest practices, wheat yield, crop damage, and selected socioeconomic characteristics of farmers, including both categorical and numerical variables.

A cluster-based sampling design was adopted, grouping farms into homogeneous clusters defined by agro-ecological conditions, geographic location, farm size, crop characteristics, and technology use.

Data were collected directly from wheat farmers through computer-assisted personal interviews (CAPI). The Egyptian component of the survey covers wheat-producing farms across multiple governorates, reflecting diverse production environments and farming practices. No sampling weights are applied, as the dataset does not provide survey weights.

2.1.2 Missing Values and Data Cleaning:

Most variables in the dataset are complete, with missing values present in only a small number of variables. Upon review, several missing entries were found to be inapplicable responses rather than true missing data and were recoded accordingly. Additionally, several categorical variables with complex or sparse response categories were recoded into broader, more interpretable groups, including variables related to crop damage reasons, input sources, weather challenges, and crop protection practices. Following these steps, the dataset was considered clean and ready for descriptive and inferential analyses, yielding a final sample of 392 observations.

2.2 Data Limitations:

While the FAO Good Growth Plan dataset provides a rich source of farm-level information on wheat production in Egypt, several limitations should be acknowledged when interpreting the findings of this study. **First**, the dataset covers a single agricultural season (2022–2023), which prevents any assessment of trends over time or causal inference, as observed associations between farm practices and outcomes may reflect seasonal conditions particular to that year rather than stable, long-term relationships. **Second**, key variables including yield estimates, cost figures, post-harvest loss percentages, and profitability status were collected through farmer interviews rather than direct measurement, meaning that self-reported values are inherently subject to recall bias and social desirability bias, which may introduce measurement error into the analysis and potentially produce influential observations that disproportionately affect model estimates. **Third**, although the dataset includes a broad range of agronomic and management variables, several potentially important determinants of wheat productivity and profitability are not captured, including farmer education level, years of farming experience, access to credit, land tenure arrangements, market distance, and extension service quality, all of which may confound or mediate the relationships estimated in the models. **Finally**, while the survey covers farms across multiple governorates, governorate-level identifiers or precise geographic coordinates are not available in the public microdata, which limits the ability to control for spatial heterogeneity in soil quality, climate conditions, and infrastructure access factors known to influence wheat outcomes in Egypt. Despite these limitations, the dataset remains one of the most comprehensive publicly available sources of farm-level wheat data for Egypt, and the analytical approach adopted in this study is designed to provide reliable findings while clearly acknowledging the study’s limitations and scope.

2.3 Analytical Framework:

This study aims to identify the key determinants of wheat productivity, post-harvest losses, profitability, and farmer decision-making in Egypt. Prior to the main empirical analysis, the dataset is explored using descriptive statistics to summarize patterns in input use, yield, post-harvest losses, revenues, and profitability. In addition, correlation analysis is conducted to examine preliminary associations between agronomic practices and key outcome variables. These exploratory steps are intended to inform model specification and support interpretation rather than establish causal relationships; the empirical analysis employs a set of complementary econometric techniques. Quantile Regression model is used to estimate the effects of agronomic, environmental, and input-related factors on wheat yield across different points of the outcome distribution, allowing for heterogeneity between low- and high-performing farms. Logistic Regression models are applied to analyze binary outcomes, including post-harvest losses, farm profitability and farmer decision-making variables such as seed-saving behavior.

To further capture the multidimensional structure of farm-level data, Factor Analysis of Mixed Data is employed to construct latent dimensions that summarize correlated quantitative and qualitative variables, including input use intensity, safety practices, and post-harvest management behavior. This approach enables dimensionality reduction while preserving the underlying structure of relationships across variables.

All statistical analyses are conducted using R to ensure reproducibility and robustness of results. Furthermore, AI-assisted tools such as ChatGPT, Claude, Gemini, and Codex were utilized to support writing refinement, assist in identifying relevant academic references, and provide guidance in the development and debugging of certain analytical code segments. These tools were used solely for language enhancement, literature support, and coding assistance, while all methodological decisions, analytical interpretations, and final code implementation remain the responsibility of the researchers.

2.4 Methodology:

2.4.1 Quantile Regression

Quantile Regression (QR), introduced by *Koenker and Bassett (1978)*, is a statistical technique for estimating the conditional median and other quantiles of the response variable. Unlike Ordinary Least Squares (OLS), which focuses on the conditional mean, QR provides a more comprehensive view of the relationship between variables across the entire distribution.

A complete theoretical and practical treatment of this methodology is provided in *Roger Koenker's* seminal book, *Quantile Regression (2005)*, which details the transition from traditional mean-based analysis to the robust framework of conditional quantiles. In this section, we will provide a summary of this technique.

OLS minimizes the sum of squared residuals to model the conditional mean of a response variable given a set of predictors *Greene (2003)*, while QR minimizes a sum of asymmetrically weighted absolute residuals. This helps in understanding how predictors influence different quantiles of the response variable. Moreover, it is particularly useful when the data is skewed, contains outliers, or exhibits heteroscedasticity (where the variance varies across the data).

QR does not partition the data, as all observations are used to estimate the parameters for every quantile using different weights.

2.4.1.1 Mathematical Form

For a given quantile $\tau \in (0,1)$, The quantile regression model is expressed as:

$$QY/X(\tau) = X\beta\tau + \varepsilon\tau$$

Where:

- $QY/X(\tau)$: The conditional τ_{th} quantile of the dependent variable Y given X.
- $\beta\tau$: The vector of coefficients specific to the τ_{th} quantile.
- $\varepsilon\tau$: The error term.

The Loss Function

Quantile regression coefficients are found by minimizing the Asymmetric Loss Function (also called the "Check Function" or "Tilted Absolute Value Function"),

denoted as ρ_τ :

$$\min_{\beta} \sum_{i=1}^n \rho_\tau(y_i - \mathbf{x}_i^T \beta)$$

Where:

$$\rho_\tau(u) = \begin{cases} \tau u & \text{if } u \geq 0 \\ (\tau - 1)u & \text{if } u < 0 \end{cases}$$

2.4.1.2 Quantile Regression Assumptions

1. **Linearity in Parameters:** The specified quantile of Y is a linear function of the parameters β .
2. **Independence of Observations.**
3. **No Perfect Multicollinearity.**
4. **Sufficient Sample Size:** For the estimation of the median ($\tau = 0.50$), a minimum sample size of 130 is generally required to achieve sufficient statistical power to detect moderate effect sizes *Green (1991)*. However, when the analysis extends to the tails of the distribution, such as the 10th or 90th percentiles, the requirement increases to at least 250 observations to ensure the stability of the estimates and reliable asymptotic inference *Shao and Wang (2009)*. These thresholds are critical for maintaining the accuracy of the standard errors and the validity of the associated hypothesis tests.
5. **No Distributional Assumption on Errors:** Unlike OLS, quantile regression does NOT require errors to be normally distributed, homoscedastic, or symmetric. That is why is a semi parametric technique.

2.4.1.3 Model's Goodness of Fit and Model Diagnostics

1. **Pseudo R^2 :** For a specific quantile τ , the Pseudo R^2 is calculated by comparing the sum of weighted absolute residuals of the full model (V_1) to a null (intercept only) model (V_0):

$$R^1(\tau) = 1 - \frac{V_1(\tau)}{V_0(\tau)}$$

The higher the value, the better the fit.

2. **Quantile residual plots:** A well-specified model should produce residuals that are symmetrically distributed around zero at the chosen quantile, with no systematic patterns when plotted against fitted values or predictors.
3. **Coverage Check:** To verify that the quantile estimates are correctly calibrated, the proportion of observations falling below each fitted quantile was examined. A well-specified model should produce proportions close to the nominal quantile levels, approximately 10% below $\tau = 0.1$, 50% below $\tau = 0.5$, and 90% below $\tau = 0.9$.
4. **Khmaladze test:** This test evaluates whether the estimated conditional quantile function is correctly specified across all quantile levels jointly, forming a global model diagnostic.

H_0 : The conditional quantile function is correctly specified

H_1 : The conditional quantile function is mis-specified

Fail to reject H_0 “The linear quantile regression model is adequate across the full distribution”.

Reject H_0 “The model is mis-specified”. We may need to consider nonlinear terms, additional covariates, or a different functional form.

2.4.2 Binary Logistic Regression

Logistic regression (LR) is a generalized linear modelling technique used to examine the relationship between a binary dependent variable and one or more independent variables *Hosmer et al (2013)*. The independent variables, also referred to as predictors or covariates, may be either continuous or categorical in nature. The main purpose of the model is to estimate the probability of occurrence of a specific event, where the dependent variable is coded as 1 to indicate the presence of the event (success) and 0 to indicate its absence (failure).

Unlike linear regression, logistic regression does not model the dependent variable directly. Instead, it models the log-odds (logit) of the probability of the event occurring. This transformation ensures that predicted values remain within the interval [0,1] and allows for a nonlinear relationship between the predictors and the outcome.

As a second step, a stepwise regression model combining both forward and backward selection was applied to identify a more efficient and predictive model after the full model revealed some insignificant coefficients. Stepwise logistics regression is a variable selection method that combines forward selection and backward elimination, using statistical criteria such as the Akaike Information Criterion (AIC) to iteratively add or remove predictors, with the goal of identifying the most efficient and statistically sound model.

2.4.2.1 The general form of the binary logistic regression model is given by *Agresti (2013)*:

$$\text{Logit}(\pi) = \alpha + \sum_{i=1}^k \beta_i X_i$$

where:

- $\pi(X_i)$: Probability of response (success or failure)
- α : is the intercept term
- β_i : are the regression coefficients associated with the predictors
- X_i : represents the independent variables
- $\text{logit}(\pi)$: log odds ratio

2.4.2.2 Assumptions of the Logistic Regression Model

For valid inference and interpretation, several assumptions are typically considered in logistic regression analysis:

- 1. Linearity in the logit for continuous variables:** While logistic regression does not require a linear relationship between predictors and the outcome, it assumes a linear relationship between continuous independent variables and the logit transformation of the dependent variable. This assumption is commonly assessed using diagnostic techniques such as the Box–Tidwell test or graphical methods.
- 2. No multicollinearity among predictors:** Independent variables should not be highly correlated with each other, as multicollinearity can inflate standard errors and reduce the reliability of coefficient estimates. This is typically assessed using the Generalized Variance Inflation Factor (GVIF). The GVIF measures the extent to which the variance of a regression coefficient is inflated due to correlation with other predictors in the model. GVIF values range from 1, indicating no multicollinearity, to infinity, indicating perfect multicollinearity. In practice, a GVIF value greater than 10 is often considered indicative of a potential multicollinearity problem that may affect the stability and interpretability of the model estimates.

2.4.2.3 Parameter estimation

The regression coefficients $\beta_0, \beta_1, \dots, \beta_k$ in a binary logistic regression model are estimated using the Maximum Likelihood Estimation (MLE) method. This approach identifies the parameter values that maximize the likelihood of observing the given sample data.

2.4.2.4 Model Fit and Evaluation

The adequacy and predictive performance of a binary logistic regression model are evaluated using several diagnostic tools that assess both model fit and predictive power.

1. Goodness-of-fit tests

There are different methods to evaluate the goodness of fit of the model or to know if the model fits the data well:

a. Pearson Chi-Square Test and Likelihood Ratio Test (G^2)

The Likelihood Ratio Test compares the fitted logistic regression model with the null model containing only the intercept.

The hypotheses are:

H_0 : All regression coefficients are equal to zero

H_1 : At least one regression coefficient is not equal to zero

A significant p-value ($p < 0.05$) indicates that the fitted model explains the dependent variable better than the null model.

b. Hosmer–Lemeshow Test

The Hosmer–Lemeshow test was used to examine the agreement between observed and predicted values.

The hypotheses are:

H_0 : The model fits the data well

H_1 : The model does not fit the data well

A non-significant p-value ($p > 0.05$) indicates good model fit and suggests no significant difference between observed and predicted values.

2. Classification accuracy and confusion matrix

A confusion matrix (classification table) is used to evaluate the predictive performance of the model. It summarizes the number of correctly and incorrectly classified observations and provides key performance measures, including:

- **Sensitivity (True Positive Rate):** proportion of actual positives correctly identified.
- **Specificity (True Negative Rate):** proportion of actual negatives correctly identified.
- **Overall accuracy:** proportion of correctly classified observations.

3. ROC curve and AUC

The Receiver Operating Characteristic (ROC) curve is used to evaluate the model's ability to discriminate between the two outcome classes by plotting sensitivity against 1-specificity across different classification thresholds. The Area Under the Curve (AUC) provides a single summary measure of model performance, where values above 0.5 indicate better-than-random classification ability, and values closer to 1 indicate excellent predictive performance *Fawcett (2006)*.

2.4.3 Factor Analysis of Mixed Data

Factor Analysis of Mixed Data (FAMD) is a data reduction method dedicated to analyzing a dataset containing both quantitative and qualitative variables and to examine the underlying structure among variables of different types *Pagès (2014); Escofier & Pagès (1994)*.

FAMD combines multiple techniques, such as Principal Component Analysis (PCA) for numerical variables and Multiple Correspondence Analysis (MCA) for categorical variables. The integration of these methods allows FAMD to effectively capture the relationships between variables and reveal underlying patterns in mixed datasets.

2.4.3.1 Factor Analysis of Mixed Data Assumptions:

1. **Correlation:** There should be underlying correlations between numerical variables and association between categorical ones.
2. **Linearity (for Quantitative Variables):** The relationships among quantitative variables are assumed to be approximately linear.
3. **No Perfect Multicollinearity.**
4. **No Missing Values.**
5. **Sample Size:** A common rule of thumb is to have at least 5-10 observations per variable to ensure stable factor loadings.

2.4.3.2 Number of Dimensions to Retain:

- **Kaiser's Rule (Eigenvalue >1):** Only dimensions whose eigenvalue exceeds 1 are retained, as they explain more variance than a single original variable.
- **Scree Plot (Elbow Method):** The number of factors to retain is chosen as the number of factors before the start of the leveling off *Cattell (1966)*.
- **Cumulative Variance Explained:** Retaining enough dimensions to account for a percentage of total variance, depending on the complexity of the research domain.

2.4.3.3 Goodness of Fit in Factor Analysis of Mixed Data:

1. **Percentage of Variance Explained.**
2. **Eigenvalues:** Eigenvalues reflect the amount of variance captured by each dimension. Dimensions with eigenvalues greater than 1 (Kaiser's Rule) are considered meaningful, as they each explain more variance than a single standardized original variable. A sharp drop in eigenvalue magnitude, visible in the scree plot, further confirms the appropriate number of dimensions to retain.
3. **Contributions of Variables:** The contribution of each variable to a given dimension indicates how much that variable drives the formation of the factor. Variables with high contributions are well represented by the dimension, suggesting a good fit between the model and the data structure. Contributions are expressed as percentages, where the sum across all variables for a given dimension equals 100%.
4. **Correlation Ratios and Squared Correlations:** For quantitative variables, the squared correlation between each variable and the extracted dimension is used to assess fit. For qualitative variables, the correlation ratio measures the association between the categorical variable and the dimension. Higher values for both indicate a stronger and more meaningful relationship between the variables and the retained factors.

Chapter (3)

Data Analysis

3.1 Data Description:

To ensure comparability across farms of different sizes, numerical variables were standardized on a per-hectare basis. Total cultivated land was calculated as the sum of permanent and temporary land. Key variables, including yield, revenue, estimated revenue, and input costs, were then expressed on a per-hectare basis by dividing each by total cultivated land. This transformation allows for a clearer assessment of productivity and efficiency by focusing on relative performance rather than absolute values, making comparisons across farms more meaningful and aligned with standard agricultural measures.

To further assess the quality of the numerical data, a boxplot was used to detect the distribution and the potential outliers in the main quantitative variable, yield per hectare (yield_ha).

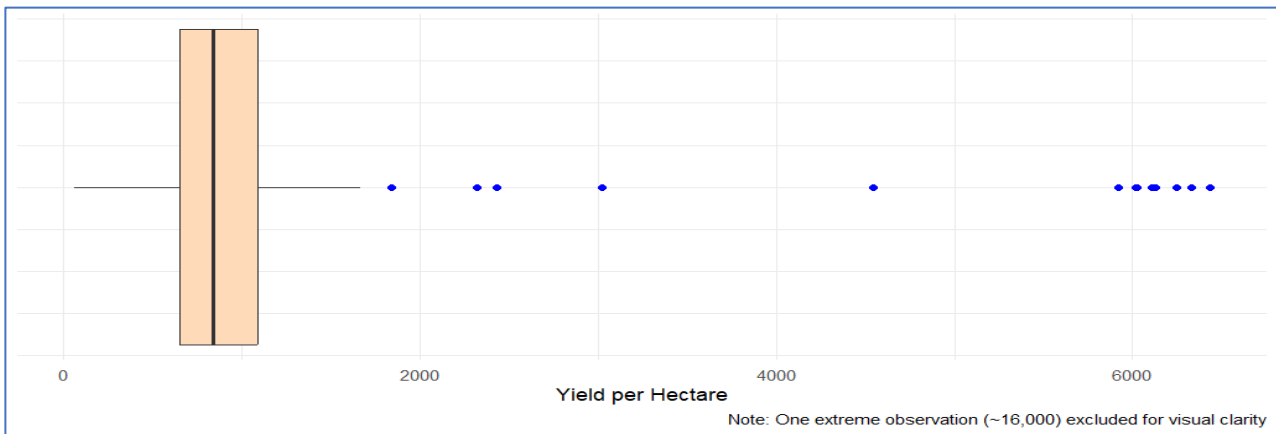


Figure 3.1: Boxplots of yield per hectare

Figure 3.1 shows the boxplot of yield per hectare. The distribution is clearly right-skewed, with the majority of farms concentrated at relatively low yield levels, as indicated by the compact interquartile range on the left side of the plot. Several high-value observations appear beyond the upper whisker, confirming the presence of outliers in the upper tail of the distribution. One extreme observation was excluded from the figure for visual clarity but remains in the dataset for analysis.

Table 3.1 Relationship between Yield and Numerical Variables

Variable	Correlation (r)	p-value
Fertilizer Cost	0.561	<0.001
Pesticide Cost	0.574	<0.001
Seed Cost	0.566	<0.001
Damaged Percentage	0.096	0.057
Fallow Land	-0.218	<0.001

Table 3.1 represents the results of the Pearson correlation coefficients. The results show a moderate positive significant relationship between yield per hectare and key input costs, including fertilizer, pesticide, and seed costs, suggesting that higher investment in agricultural inputs is associated with improved productivity.

Conversely, damaged percentage shows a very weak positive correlation with yield, indicating a limited linear relationship with productivity in this sample, while total fallow land is weakly negatively correlated with yield, suggesting that larger shares of unused land are slightly associated with lower productivity per hectare.

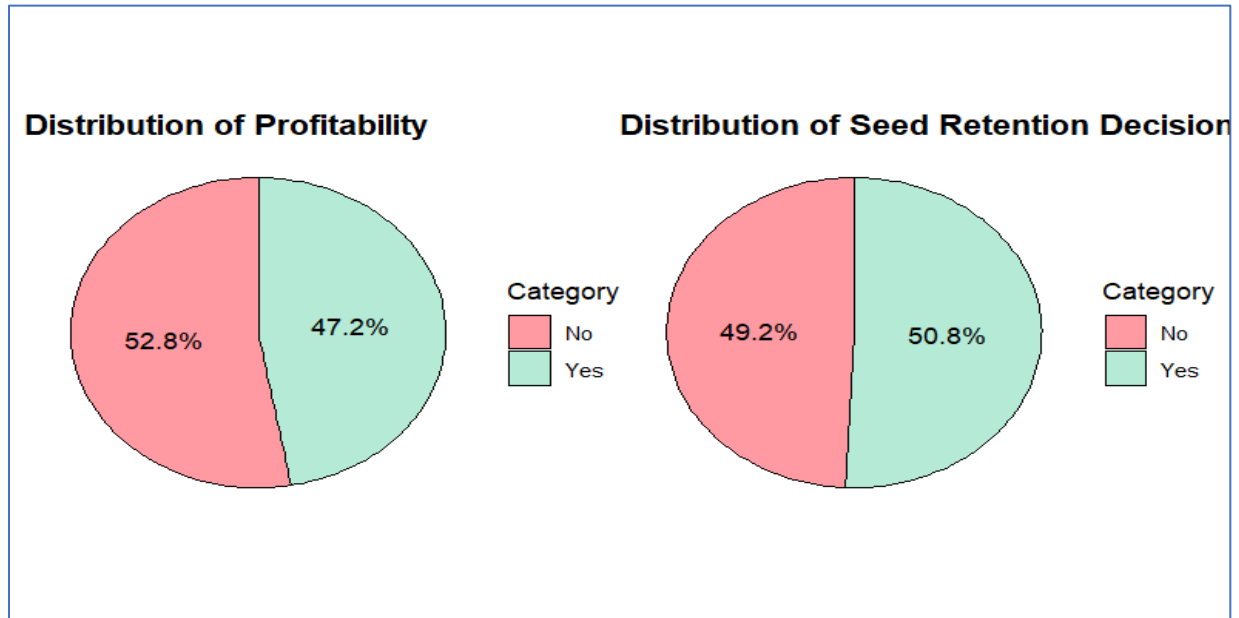


Figure 3.2: Distribution of Profitability and Seed Retention Decisions

Figure 3.2 shows that both profitability status and seed retention decisions are relatively evenly distributed among farmers. This indicates the absence of a dominant pattern in either variable.

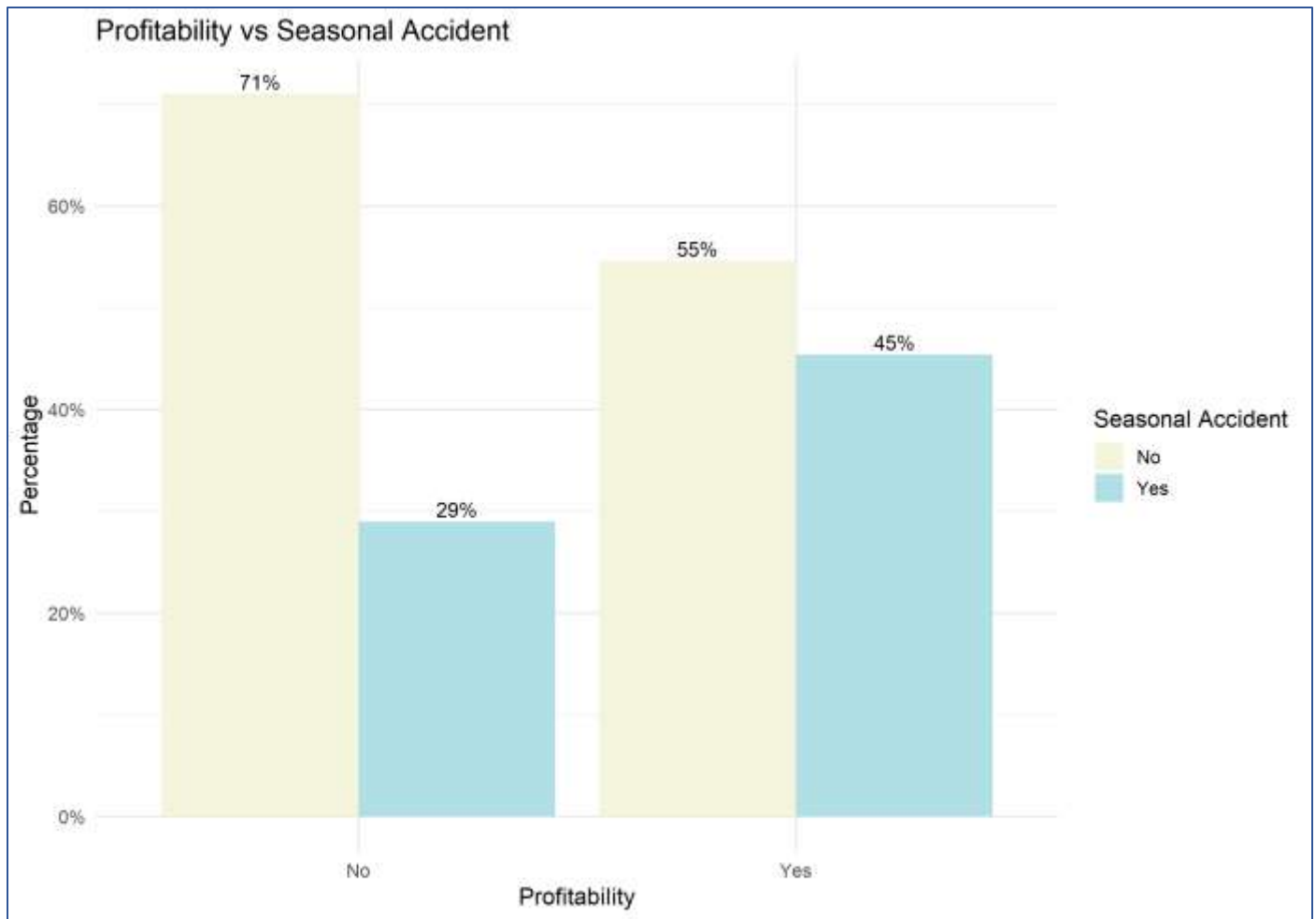


Figure 3.3: Seasonal Accidents & Profitability

Figure 3.3 shows that profitable farms reported a similar proportion of seasonal accidents compared to non-profitable farms, with only marginal differences between the two groups. The association between seasonal accidents and profitability remains weak ($\tau^4 = 0.029$, $p\text{-value} = 0.001108$), indicating that profitability status alone is not a strong differentiator of accident occurrence.

⁴ Goodman-Kruskal's Tau is an asymmetric measure of association between two nominal categorical variables. It quantifies how much the knowledge of one variable reduces the prediction error of another variable.

3.2 Statistical Modeling and Results:

3.2.1 The Determinants of Wheat Yield in Egypt

The Key Role of Water Management, Input Costs, Crop Protection, and Advisory Services in Shaping Yield Performance among Egyptian Farmers:

The initial dataset was screened for leverage points using the hat matrix to identify observations with unusual combinations of predictor variables that could disproportionately influence the model.

Observations with hat values exceeding the threshold of $2p/n$ were flagged as high-leverage points, following the criteria established by *Belsley, Kuh, and Welsch (1980)*. Accordingly, 15 observations were excluded from the analysis, yielding a final sample size of 377 observations.

3.2.1.1 Model Selection and Validation

To identify the main factors influencing wheat yield, we started with (OLS) linear regression. However, diagnostic testing revealed clear violations of its core assumptions, including non-normality of residuals detected ($p < 0.001$), heteroscedasticity, and influential outliers that biased the estimates. Since OLS models are the conditional mean, these violations were enough to undermine the reliability of the results. We then tried a Gamma generalized linear model (GLM), which seemed better suited to the skewed nature of yield data *McCullagh & Nelder (1989)*. Diagnostic tests revealed a poor distributional fit, significant overdispersion ($p = 0.008$) that made standard errors unreliable, and evidence of non-linear patterns and extreme values the model could not adequately handle ($p < 0.0001$).

We therefore adopted quantile regression to identify the determinants of wheat yield. Unlike the previous methods, quantile regression requires no distributional assumptions. It is robust to outliers and captures how predictors relate to wheat yield across its entire distribution, not just at the mean. For farm-level data characterized by variability and extremes, this flexibility made it the most appropriate choice.

3.2.1.2 Testing the Assumptions of the Model

Table 3.2: GVIF for Model Predictors

Variable	GVIF	Variable	GVIF
Water source	2.848	Seed cost ha	3.621
Soil test	1.037	Agro advice	1.041
Spraying advice	1.052	Damaged pct ha	1.995
Mixture protection	1.081	Spray guide	1.060
Adv weather	1.044	Get products	1.069
Ferti cost ha	2.699	Total fallow land	1.912
Pesti cost ha	2.411	Input from	1.197

Table 3.2 presents the GVIF values for all model predictors. All variables fall well below the acceptable threshold of 10, indicating no multicollinearity concerns.

3.2.1.3 Model's Goodness of Fit and Model Diagnostics

1. Pseudo R^2 :

Table 3.3 Pseudo R^2 Values across Selected Quantiles

Tau (τ)	0.1	0.5	0.9
Pseudo R^2	0.606	0.579	0.735

The Pseudo R^2 values indicate that the model's explanatory power varies significantly across different segments of the productivity distribution, demonstrating that the determinants of wheat yield are more clearly defined at the extremes than at the center.

The model achieves its highest fit at the 90th quantile ($\tau = 0.9$), with a Pseudo R^2 of 0.7354, indicating that the included technical inputs and management practices account for approximately 74% of the variation in output among high-yield farmers. Similarly, the model performs strongly at the 10th quantile ($\tau = 0.1$), with a Pseudo R^2 of 0.6061, suggesting that the selected variables effectively capture the conditions characterizing low-production environments. The slight decrease in fit at the median ($\tau = 0.5$) suggests that mid-level farmers are influenced by a broader mix of factors, some of which may not be fully captured by the included variables.

Overall, the Pseudo R^2 values confirm that the chosen variables provide meaningful explanatory power across the productivity distribution, with particularly strong explanatory performance at both tails.

2. Quantile residual plots:

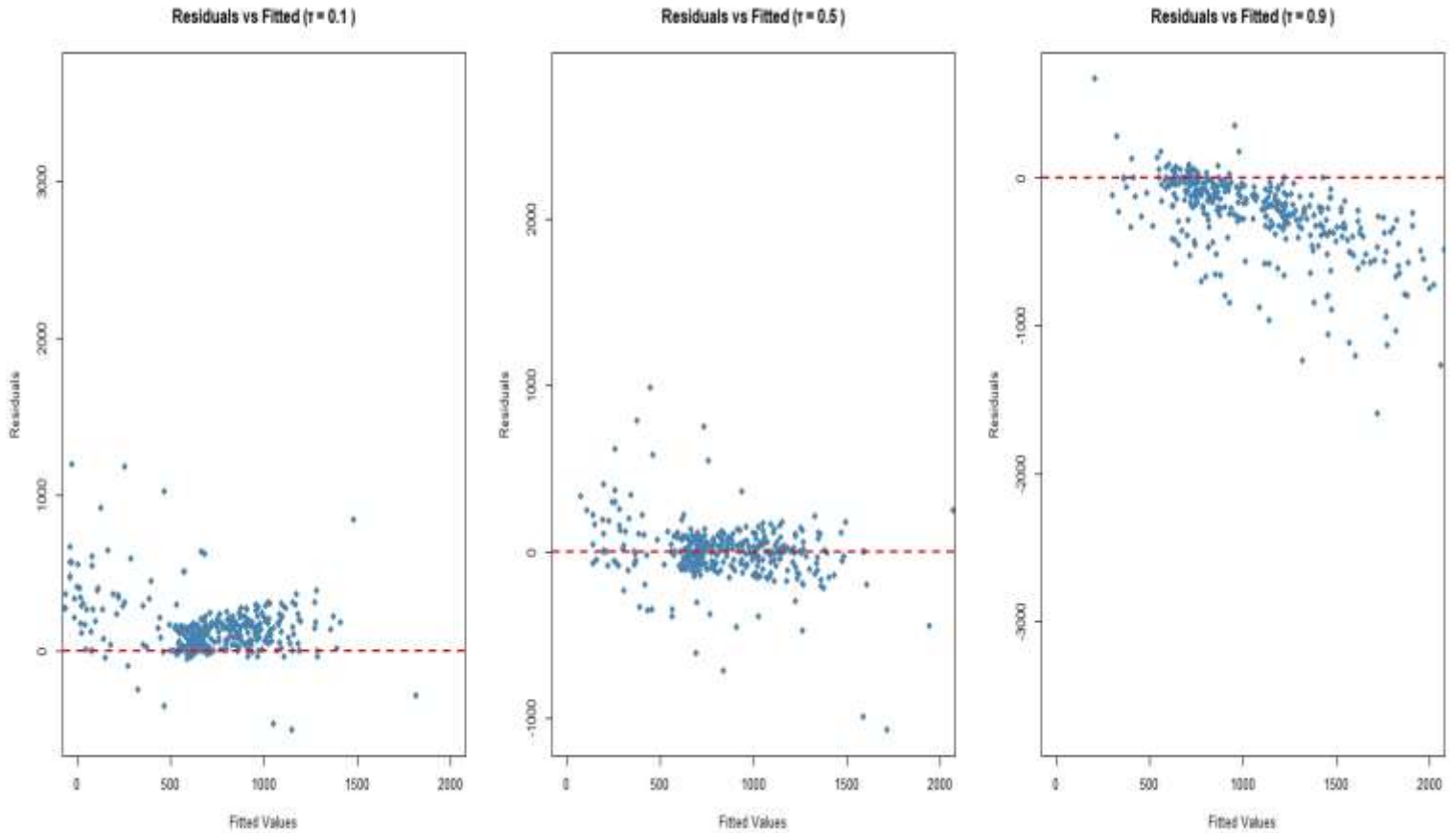


Figure 3.4: Residuals vs Fitted Values across Selected Quantiles ($\tau = 0.1, 0.5, 0.9$)

The residual plots displayed in Figure 3.4 provide supporting evidence of model adequacy across the distribution of wheat yield. As expected by construction, the 10th quantile plot shows residuals concentrated above the zero line, while the 90th quantile plot shows them predominantly below it. This directional pattern is a direct consequence of how quantile regression works mathematically when fitting a model at the 10th quantile ($\tau = 0.1$), the model draws a line such that only 10% of observations fall below it, meaning 90% of residuals will naturally be positive and sit above the zero line. The opposite holds at the 90th quantile ($\tau = 0.9$), where the model draws a line that 90% of observations fall below, resulting in most residuals being negative. At the median ($\tau = 0.5$), the model splits observations equally above and below the fitted line, producing residuals that are approximately symmetrically scattered around zero with no clear systematic pattern, providing the strongest evidence of adequate model fit at the center of the distribution. These directional patterns are not a flaw or a sign of poor fit; they are precisely how a correctly specified quantile regression model is supposed to behave, confirming that the model is

functioning as intended. Overall, the residual plots support the conclusion that the quantile regression model provides a reasonable and adequate fit across the full distribution of wheat yield.

3. Coverage Check:

The coverage checks provide further evidence of model calibration across the distribution. At ($\tau = 0.1$), approximately 13% of observations fell below the fitted quantile, slightly higher than the expected 10% but remaining reasonably close. At the median ($\tau = 0.5$), 47.2% of observations fell below the fitted value, marginally below the expected 50%, and well within an acceptable range. At ($\tau = 0.9$), 88.1% of observations fell below the fitted quantile, slightly lower than the expected 90% but still a close approximation.

Overall, these results confirm that the quantile regression model is producing well-calibrated estimates across the distribution of wheat yield, with no systematic or alarming departures from the expected proportions.

4. Khmaladze test:

The Khmaladze test indicated no significant evidence of model misspecification ($p = 1$), confirming that the linear quantile regression model is correctly specified across the full distribution of wheat yield.

This provides strong global evidence of model adequacy, complementing the satisfactory Pseudo R^2 values, well-calibrated coverage checks results, and acceptable residual plots reported previously.

3.2.1.4 Interpreting the Results of Quantile Regression

Table 3.4: Quantile Regression Coefficient Estimates for Wheat Yield per Hectare at the 10th, 50th, and 90th Quantiles

Characteristic	10 th Quantile Beta	10 th Quantile p-value	50 th Quantile Beta	50 th Quantile p-value	90 th Quantile Beta	90 th Quantile p-value
Water Source						
Irrigated	—	—	—	—	—	—
Rain-fed	278	0.02*	-5.9	>0.9	-364	<0.001***
Soil Test						
No	—	—	—	—	—	—
Yes	-2.5	0.8	-21	0.2	28	0.3

Characteristic	10 th Quantile Beta	10 th Quantile p-value	50 th Quantile Beta	50 th Quantile p-value	90 th Quantile Beta	90 th Quantile p-value
Spraying Advice						
No	—	—	—	—	—	—
Yes	2.5	0.7	-23	0.082	35	0.2
Mixture Protection						
L	-2.7	0.7	21	0.2	-2.8	>0.9
Q	6.7	0.3	-20	0.15	-9.2	0.7
Adv. Weather						
Drought	—	—	—	—	—	—
Excessive Heat	3.1	0.7	14	0.3	42	0.14
Ferti. Cost/ha	-0.08	>0.9	0.38	0.5	0.93	0.2
Pesti. Cost/ha	35	<0.001***	44	<0.001***	63	<0.001***
Seed Cost/ha	2.9	0.5	-0.52	0.9	7.6	<0.001***
Agro Advice						
No	—	—	—	—	—	—
Yes	5.7	0.4	9.6	0.5	-19	0.5
Damaged %/ha	7.1	0.2	-9.8	0.4	-56	0.003**
Spray Guide						
No	—	—	—	—	—	—
Yes	-21	0.039*	-19	0.2	-43	0.1
Get Products						
No	—	—	—	—	—	—
Yes	-14	0.078	18	0.2	29	0.3
Total Fallow Land	1.1	0.4	-1.7	0.3	-9	0.008**
Input From						
Formal	—	—	—	—	—	—
Informal	-12	0.4	16	0.4	14	0.7
Market	-8.5	0.4	4.6	0.8	2.7	>0.9
Other	0.37	>0.9	-4	0.9	-53	0.12

Note: * p < 0.05. ** p < 0.01. *** p < 0.001

Water Source Rain-fed farming showed a significant positive effect among low-yield farmers, confirming that access to natural rainfall is a meaningful advantage for struggling farmers. However, this effect reverses dramatically at the upper quantile, suggesting that among high-yield farmers, reliance on rain-fed sources becomes a constraint and that efficient irrigation systems are what distinguishes top

performers. This sign reversal is a particularly important finding as it reveals that water source requirements differ fundamentally depending on a farmer's productivity level.

Soil Test Soil testing showed no significant effect across any quantile, suggesting that in this sample, soil testing alone is not a direct driver of yield variation regardless of farmer productivity level.

Spraying Advice Following label advice when preparing for spraying showed no significant effect across any quantile, suggesting that while it remains a best practice, it does not independently drive yield variation in this sample.

Pesticide Cost per Hectare This is the most consistently significant variable across all quantiles, with a strongly increasing positive effect from low to high-yield farmers. This confirms that pesticide investment is the single most universal and powerful driver of wheat yield, with its importance growing stronger as farmer productivity increases, making it the clearest and most actionable finding of this study.

Seed Cost per Hectare Seed cost showed no significant effect at the lower or median quantiles but became highly significant at the upper quantile, confirming that seed quality investment is primarily a distinguishing factor among high-yield farmers. For low and median-yield farmers, other more fundamental constraints appear to be limiting productivity before seed quality can make a meaningful difference.

Damaged Percentage per Hectare Crop damage showed a significant negative effect only among high-yield farmers, confirming that post-harvest losses are a meaningful and direct constraint on productivity at the upper end of the distribution. For low and median-yield farmers, crop damage did not show a significant effect, suggesting that other factors dominate their yield outcomes.

Spray Guide Receiving guidelines from buyer or exporter advisors showed a significant negative effect among low-yield farmers, which is a counterintuitive finding. This may reflect that buyer guidelines are typically designed for general or ideal farming conditions rather than the specific challenges that low-yield farmers face locally. Additionally, participating in buyer-guided programs often requires following rigid protocols regardless of local soil, water, or climate conditions, and may involve purchasing specific products that low-yield farmers cannot fully afford, creating financial pressure without the expected yield return. It is worth noting that these guidelines are not inherently harmful; they appear to work well for high-yield farmers who have the resources to implement them effectively. For low-yield farmers, however,

the rigid application of externally designed protocols in challenging local conditions may work against rather than in favor of their productivity, highlighting the need for locally adapted advisory programs.

Total Fallow Land Fallow land showed a significant negative effect among high-yield farmers, suggesting that, among top performers, having more uncultivated land is associated with lower yield. This likely reflects that high-performing farmers have already optimized their land use strategies, and fallow land represents an opportunity cost land that could otherwise be contributing to output is left unproductive.

The remaining variables, **Mixture Protection, Excessive Heat, Fertilizer Cost per Hectare, Access to Agricultural Products, Input Source, and Agronomic Advice** showed no significant effect across any quantile.

Overall, the determinants of wheat yield in Egypt are not uniform across the productivity distribution. Pesticide investment is the only factor that matters universally for all farmers.

Water source, seed cost, crop damage, spray guidance, and fallow land management all tell fundamentally different stories depending on where a farmer sits in the yield distribution, further confirming that a one-size-fits-all approach to agricultural policy would be inadequate for addressing the diverse needs of Egyptian wheat farmers.

While these findings highlight the heterogeneous drivers of wheat productivity at the production stage, improving overall agricultural performance also requires close attention to what happens after harvest. In many cases, gains achieved in the field can be partially or entirely offset by losses occurring during post-harvest handling, storage, and processing. Therefore, to develop a more comprehensive understanding of wheat system performance, the analysis now shifts from yield determination to the factors influencing post-harvest losses among farmers.

3.2.2 Post-Harvest Wheat Loss

The Key Biological, Environmental, and Farm Management Factors Driving High Post-Harvest Wheat Loss among Farmers:

Post-harvest wheat damage is a major agricultural challenge that affects farmer profitability and food security. Loss levels may vary according to biological pressures, environmental conditions, farm management practices, and institutional support. Factors such as pest infestations, adverse weather, water sources, spraying practices, and access to agricultural guidance may all contribute to differences in wheat loss levels among farmers.

Understanding the factors associated with different post-harvest loss levels is therefore essential for designing targeted interventions that enhance wheat preservation, improve farm profitability, and strengthen food system resilience. Identifying the conditions linked to elevated loss levels can support policymakers, extension services, and agricultural stakeholders in prioritizing risk-reduction strategies and improving post-harvest management practices.

Accordingly, this study investigates the key determinants associated with high and low post-harvest wheat loss levels among Egyptian wheat farmers using binary logistic regression due to violations of classical linear regression assumptions.

To classify post-harvest wheat loss into meaningful groups, the distribution of the continuous variable (damage per hectare) was examined using kernel density⁵ estimation.

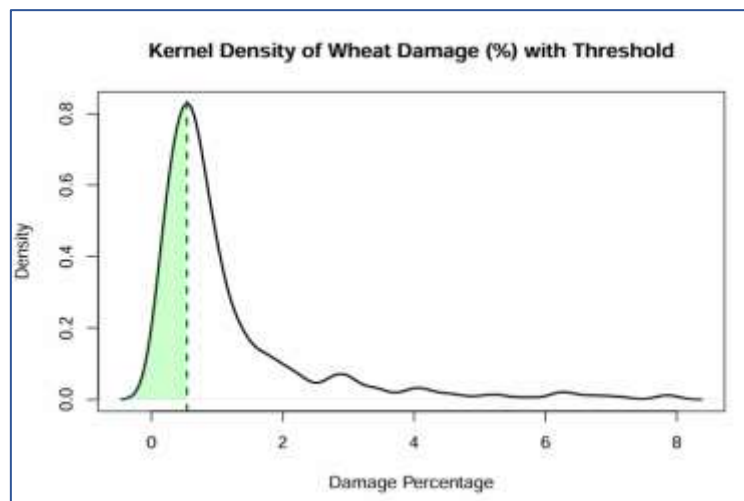


Figure 3.5: Kernel Density Distribution of Wheat Post-Harvest Loss (%)

Using the Kernel Density plot in Figure 3.5, an appropriate cutoff point was chosen to transform the post-harvest wheat loss into a binary variable. The threshold 0.542% was chosen to distinctly separate the groups. Consequently, the binary variable is assigned a value of 0, conforming to the label “**low loss**,” if the post-harvest wheat loss is less than 0.542%, otherwise, it is assigned a value of 1, corresponding to the label “**high loss**”.

⁵ Kernel density estimation is a non-parametric method used to estimate the probability distribution of a continuous variable. It produces a smooth curve representing the data distribution without assuming any specific functional form, allowing for the identification of key features such as the mode and overall shape of the distribution.

Out of the 392 observed cases, 128 farmers (32.7%) were classified in the low-loss category, while 264 farmers (67.3%) were classified in the high-loss category.

The initial full logistic regression model included a relatively large set of explanatory variables covering biological factors, environmental conditions, farm management practices, and institutional characteristics related to post-harvest wheat loss.

A stepwise regression approach based on AIC was applied. Through iterative variable selection, non-contributing variables were gradually removed while retaining those that improved model fit.

The general specification of the Final model is given by:

$$\text{logit}(\hat{\pi}_i) = \beta_0 + \beta_1(\text{pest}) + \beta_2(\text{water_source}) + \beta_3(\text{season_accident}) + \beta_4(\text{input_from})$$

3.2.2.1 Testing the Assumptions of the Binary Logistic Regression Model

Table 3.5: Adjusted GVIF for variables in the model

Variable	Adjusted GVIF
Input from	1.0109
pest	1.0212
water source	1.1538
seasonal accident	1.1314

GVIF was calculated for all four variables selected in the final regression model to check whether there exists any multicollinearity. In view of differences between predictors in terms of degrees of freedom, the adjusted GVIF statistic was considered as a measure of multicollinearity test. From Table 3.5, all the adjusted GVIF values lie well below 10, implying that no multicollinearity exists in the model.

3.2.2.2 Testing Goodness of Fit

a) Likelihood Ratio Test (G^2)

The likelihood ratio test indicates that the full model provides a significantly better fit compared to the intercept-only model ($p < 2.2e-16$), confirming that the included predictors jointly contribute to explaining variation in post-harvest wheat loss levels.

b) ROC Curve and AUC

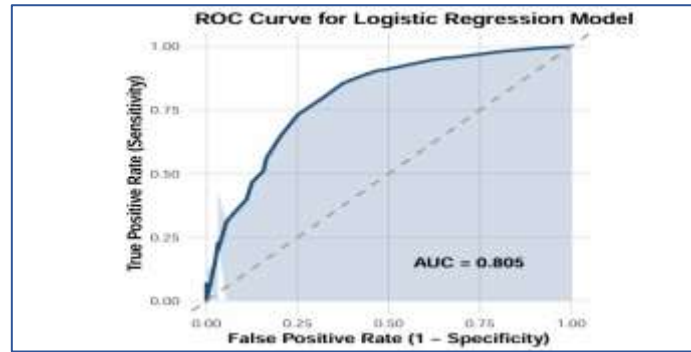


Figure 3.6: ROC Curve for the LR Model (Post-Harvest Wheat Loss)

The ROC curve indicates the model is capable of correctly distinguishing between high and low post-harvest loss cases in approximately 80% of observations.

c) Classification Table

To evaluate the predictive power of the developed model, the threshold for predicting the correct class of each farm was optimally determined through the application of the Youden Index. It defines the threshold value at which sensitivity and specificity will have maximum values by the formula:

$$J = \text{Sensitivity} + \text{Specificity} - 1$$

Using a classification threshold of 0.599, the logistic model demonstrates good predictive performance with an accuracy of 78.1%, significantly higher than the no-information benchmark. The model performs particularly well in identifying high-loss farmers (sensitivity = 0.86), although specificity is comparatively lower (0.63), indicating a tendency to better capture risk cases than non-risk cases. Overall, the results suggest that the model has strong discriminatory power and is particularly effective for identifying farmers exposed to high post-harvest losses.

3.2.2.3 Interpreting the Results of the Binary Logistic Model

Table 3.6 presents the results of the logistic regression model for post-harvest wheat loss.

Table 3.6: Results of fitting binary LR model (Post-Harvest Wheat Loss)

Variable	Odds Ratio	Std. Error	p-value
Intercept	0.616	0.284	0.089 ·
pest			
Rodents	0.676	0.257	0.127
Water Source			
Rain-fed	9.356	0.315	1.27e-12 ***
Seasonal Accident			
Yes	2.0219	0.318	0.027 **
Input from			
Informal	1.0951	0.318	0.8197
Market	0.3167	0.317	0.0003 ***
Other	1.0629	0.419	0.884

Note: p < 0.1. **p < 0.01. ***p < 0.001

Water Source: Water source is the most statistically significant predictor in the model. Farmers relying on rain-fed agriculture have approximately 9.36 times higher odds of experiencing high post-harvest wheat loss compared to farmers using irrigated water sources, holding all other variables constant. This finding suggests that dependence on natural rainfall increases farmers' vulnerability to unfavorable environmental conditions, moisture variability, and production instability, all of which may contribute to higher post-harvest losses.

Seasonal Accident: Experiencing a seasonal accident shows a positive and statistically significant association with high post-harvest wheat loss. Farmers who experienced seasonal accidents have approximately 2.02 times higher odds of suffering high post-harvest losses compared to farmers who did not experience such accidents, holding other variables constant. This result indicates that unexpected field disturbances and adverse seasonal conditions can substantially increase production and storage-related losses.

Input from: The source from which farmers obtain agricultural inputs shows mixed effects on post-harvest loss levels. Farmers sourcing inputs from market channels have approximately 68.3% lower odds of experiencing high post-harvest wheat loss compared to farmers relying on formal input sources, holding other variables constant. This effect is highly statistically significant and may indicate that market-based channels provide more accessible, timely, or suitable production inputs that help farmers reduce losses.

Overall, the logistic regression results suggest that environmental exposure and production-related conditions are the primary drivers of post-harvest wheat loss among farmers. Reliance on rain-fed agriculture and exposure to seasonal accidents significantly increase the likelihood of high losses, while market-based input sourcing appears to reduce loss risk. These findings emphasize the importance of strengthening irrigation systems, improving farmers' resilience to environmental shocks, and enhancing access to reliable agricultural inputs to mitigate post-harvest wheat losses.

While reducing post-harvest losses is essential for improving overall system efficiency, farmers' economic outcomes ultimately depend on how production costs and resource allocation translate into profitability. Even in the presence of yield improvements and reduced losses, variations in cost structures can significantly influence net returns and financial sustainability at the farm level. Therefore, the analysis now turns to examining how production costs and their composition affect farm profitability among wheat producers.

3.2.3 Farm Profitability

The Effect of Production Costs and Cost Structure on Farm Profitability:

To analyze the effect of production costs and cost structure on farm profitability, a binary logistic regression model was applied. The dependent variable was based on the classification of farms, where 1 was assigned if the farm was profitable and 0 otherwise. Out of the 392 cases observed, 185 farms (47.2%) were classified as profitable, while 207 farms (52.8%) were unprofitable.

Since many predictors were available, the model-building process followed a sequential approach. The first model fitted included all seven individual cost components per hectare alongside all non-cost predictors, resulting in 26 predictors. However, this model suffered from severe multicollinearity among cost variables and a low Events Per Predictor ratio of 7.1⁶, indicating overfitting.

To resolve this, the seven cost components were replaced by a single aggregated variable⁷, the natural log of total cost per hectare, reducing predictors to 20 and resolving the multicollinearity issue. A bidirectional stepwise selection procedure was then applied using AIC as the selection criterion, reducing the model

⁶ Events per predictor (EPP) refers to the number of outcome events (profitable farms) per estimated predictor in a logistic regression model. A commonly recommended minimum is 10 events per predictor to ensure stable and reliable estimates; values below this threshold may indicate potential overfitting.

⁷ The aggregated cost variable was constructed by summing all individual cost components per hectare to obtain total cost per hectare, after which a natural logarithmic transformation was applied.

further to five predictors. The substantial difference in AIC values ($\Delta AIC > 10$) provides strong evidence in favor of this model with no significant loss in discrimination ability (Burnham,2002). The DeLong test showed no significant difference between the AUCs of the two models ($p = 0.3084$); therefore, it was selected as the final model

The general specification of the Final model is given by:

$$\text{logit}(\hat{\pi}_i) = \beta_0 + \beta_1(\text{log_total_cost}) + \beta_2(\text{water_source}) + \beta_3(\text{yield_ha}) + \beta_4(\text{pest}) + \beta_5(\text{season_accident})$$

- Log of total production cost per hectare (log_total_cost)
- Water source type (water_source)
- Yield per hectare (yield_ha)
- Primary pest type (pest)
- Occurrence of a seasonal accident (season_accident)

3.2.3.1 Testing the Assumptions of the Model

1. Linearity in the Logit for Continuous Variables

Table 3.7: Box–Tidwell Test for Linearity of Continuous Variables (Profitability Prediction)

Variable	Estimate	Std. Error	z value	p-value
(Intercept)	-0.103	0.424	-0.242	0.809
Yield per Hectare	0.281	0.469	0.601	0.548
Interact Yield per Hectare	-0.287	0.248	-1.157	0.247

As presented in Table 3.7, the interaction term was statistically insignificant ($p > 0.05$), confirming that the relationship between the continuous predictor and the logit of farm profitability is linear, and the assumption is satisfied.

2. No Multicollinearity Among Predictors

Table 3.8: Adjusted GVIF for variables in the model (Profitability Prediction)

Variable	Adjusted GVIF
Log Total Cost per Hectare	1.947
Water Source	1.371
Yield per Hectare	1.840
pest	1.013
Seasonal Accident	1.257

GVIF was calculated for all five variables selected in the final regression model to check whether there exists any multicollinearity. In view of differences between predictors in terms of degrees of freedom, the adjusted GVIF statistic was considered as a measure of multicollinearity test. From Table 3.8, all the adjusted GVIF values lie well below 10, implying that no multicollinearity exists in the model.

3.2.3.2 Testing Goodness of Fit

a) Pearson Chi-Square Test and Likelihood Ratio Test (G^2)

The overall significance of the final model was determined based on the results of the Likelihood Ratio Test (LRT), in which the fitted model was compared with the null model that includes only the intercept. The test produced a p-value = 0.0001, suggesting that the model is statistically significant overall and fits the data better than the null model. This indicates that the five selected predictors play an important role in explaining the likelihood of farm profitability.

b) Hosmer-Lemeshow Test

The Hosmer-Lemeshow goodness-of-fit test was applied to examine the consistency of the prediction made by the model and the actual data observed across various risk levels. Since the model contains continuous predictors, this test is particularly appropriate. The test was conducted using both $g = 10$ and $g = 5$ groups to ensure robustness, as the result can be sensitive to the choice of grouping. Both groupings yielded p-values well above 0.05, indicating that there is no significant difference between the observed and predicted values, and that the model fits the data well.

c) ROC Curve and AUC

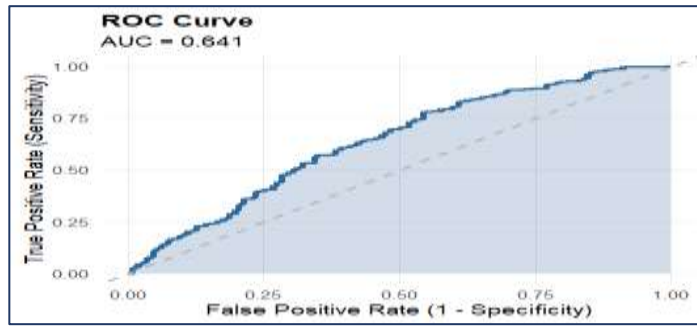


Figure 3.7: ROC Curve for the LR Model (Profitability Prediction)

The (ROC) curve was plotted to examine the model's capability in discriminating between the profitable and unprofitable farm operations across all possible classification thresholds. As shown in Figure 3.7, the ROC curve lies clearly above the diagonal reference line, which represents random classification. The (AUC) was 0.641, indicating modest discriminatory ability above chance.

d) Classification Table

The optimal threshold was selected using the Youden Index is 0.573. Using this threshold, the model classified 65.4% profitable farms as true positive (Sensitivity) and 57.0% unprofitable farms as true negative (Specificity). The overall accuracy equals 61.0%.

3.2.3.3 Interpreting the Results of the Binary Logistic Model

Table 3.9 presents the results of the final binary logistic regression model, including odds ratios, standard errors, and p-values for each predictor.

Table 3.9: Results of fitting binary LR model (Profitability Prediction)

Variable name	Odds ratio	Standard error	p-value
Intercept	0.947	0.478	0.909
Log Total Cost per Hectare	2.244	0.449	0.072
Water Source			
Rain-fed	1.529	0.282	0.132
Irrigated	.	.	.
Yield per Hectare	0.622	0.184	0.010 **
Pest Type			
Rodents	0.694	0.210	0.083
None	.	.	.
Seasonal Accident			
Yes	0.4208	0.2414	< 0.001 ***
No	.	.	.

Note: . p < 0.1. * p < 0.05. ** p < 0.01. *** p < 0.001

Log Total Cost per Hectare: The log of total production cost per hectare shows a positive association with farm profitability. Farms with higher total production costs per hectare have approximately 2.24 times higher odds of being profitable, holding other variables constant. This finding suggests that farms utilizing high levels of production inputs, including seeds, fertilizers, pesticides, farm equipment, and irrigation, are associated with higher profitability, potentially due to the greater income earned from intensive farming methods.

Yield per Hectare: Yield per hectare is a statistically significant negative predictor of farm profitability at the 5% level. A one-unit increase in yield per hectare (measured in 1,000 kg/ha) is associated with a reduction in the odds of profitability by approximately 37.9%, holding other factors constant. While this result may appear counterintuitive, it is likely explained by the fact that total production cost is already controlled by the model. Higher yields often require 38% greater input intensity, and once costs are accounted for, additional yield gains may reflect diminishing returns where marginal revenue does not fully offset the additional expenditure required to achieve them.

Pest Type: The presence of rodents as the primary pest type shows a negative association with farm profitability compared to pest-free farms, marginally significant at the 10% level. The odds of being profitable among farms affected by rodents are approximately 30.6% lower than those of pest-free farms, holding other variables constant. Rodent infestations are associated with substantial pre- and post-harvest losses, reducing the quantity of marketable output and consequently farm revenues. Given the marginal significance, this finding warrants cautious interpretation.

Seasonal Accident: Experiencing a seasonal accident is the most statistically significant predictor in the model. The odds of being profitable among farms that experienced a seasonal accident are approximately 57.9% lower compared to farms that did not, holding other variables constant.

Having established that seasonal accidents, pest pressure, and input intensity are key determinants of farm profitability, a related but distinct behavioral question emerges: how do these economic outcomes and production characteristics shape farmers' post-harvest decisions? One of the most consequential decisions a wheat farmer makes after harvest is whether to retain part of the yield as seed for the following season, a choice that directly reflects the farmer's economic confidence, input strategy, risk tolerance. To identify the key determinants of this seed retention behavior, a binary logistic regression model was estimated.

3.2.4 Seed Retention Decisions

The Determinants of Seed Retention Decisions among Egyptian Wheat Farmers:

To identify the key determinants of seed retention behavior, a binary logistic regression model was estimated. The dependent variable takes a value of 1 if the farmer retains part of the harvest as seed for the following season, and 0 otherwise.

The modelling process began with the estimation of a full model including a wide range of economic, production, and farm-level variables. These variables include yield, input costs, revenue, weather conditions, advisory access, and indicators of farm shocks.

To obtain a more parsimonious and interpretable specification, a bidirectional stepwise selection procedure based on the AIC was applied.

The general specification of the Final model is given by:

$$\text{logit}(\hat{\pi}_i) = \beta_0 + \beta_1(\text{yield_ha}) + \beta_2(\text{seed_cost_ha}) + \beta_3(\text{machi_cost_ha}) + \beta_4(\text{accident_bin})$$

3.2.4.1 Testing the Assumptions of the Model

To assess the validity of the estimated logistic regression model, several diagnostic checks were conducted, focusing on multicollinearity, linearity in the logit, and overall model fit.

1. Linearity in the Logit for Continuous Variables

Table 3.10: Box–Tidwell Test for Linearity of Continuous Variables (Seed Retention Decision)

Variable	Estimate	Std.Error	z value	p-value
Yield per Hectare	-0.557	0.631	-0.883	0.377
Seed Cost per Hectare	-1.288	1.599	-0.806	0.420
Machinery Cost per Hectare	-0.952	1.248	-0.763	0.445

Since all p-values exceed 0.05, the results indicate that the linearity assumption in the logit is satisfied for all continuous variables at the 5% significance level.

2. No Multicollinearity Among Predictors

Table 3.11: Adjusted GVIF for variables in the model (Seed Retention Decision)

Variable	Adjusted GVIF
Yield per Hectare	3.74
Seed Cost per Hectare	1.33
Machinery Cost per Hectare	3.98
Seasonal Accident	1.11

All GVIF values are below the commonly used threshold of 10, indicating that multicollinearity is not a concern. This suggests that the estimated coefficients are stable and not driven by strong correlations among the explanatory variables.

3.2.4.2 Testing Goodness of Fit

a) Pearson Chi-Square Test and Likelihood Ratio Test (G^2)

The overall significance of the final logistic regression model was assessed using the Likelihood Ratio Test, which compares the fitted model with the intercept-only model. The p-value of 0.00009 indicates that the explanatory variables jointly provide a statistically significant improvement over the intercept-only model.

b) Hosmer-Lemeshow Test

Using the Hosmer–Lemeshow Test, there is no evidence of lack of fit (p-value = 0.521), indicating that the model fits the data adequately.

c) ROC Curve and AUC

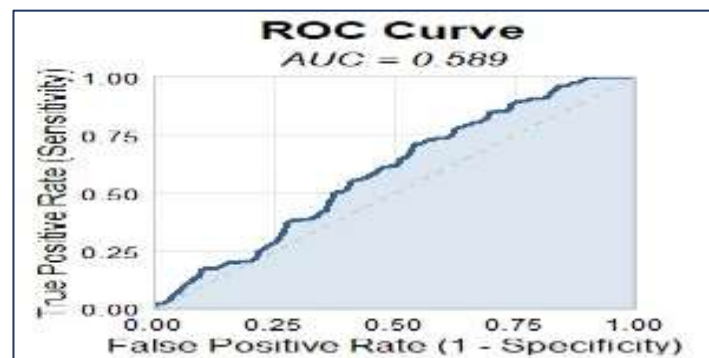


Figure 3.8: ROC Curve for the LR Model (Seed Retention Decision)

The model shows moderate predictive performance. The AUC indicates that the model has limited ability to distinguish between farmers who retain seeds and those who do not.

d) Classification Table

The predictive performance of the model was further assessed using a classification table based on a threshold of 0.495, which is determined using the Youden Index. The calculated Classification Metrics are (Accuracy = 0.586), (Sensitivity = 0.708), and (Specificity = 0.461). The classification results indicate that the model correctly classifies about 58.6% of observations. The sensitivity value shows that the model is better at identifying farmers who retain seeds than those who do not, while the lower specificity indicates weaker performance in correctly identifying non-retainers. Overall, the predictive performance of the model is moderate.

3.2.4.3 Interpreting the Results of the Binary Logistic Model

Table 3.12: Results of fitting binary LR model (Seed Retention Decision)

Variable name	Odds ratio	Standard error	p-value
Intercept	1.82	0.2583	0.0199 *
Yield per Hectare	0.391	0.5007	0.0611·
Seed Cost per Hectare	0.272	0.7610	0.0875·
Machinery Cost per Hectare	6.30	1.1414	0.1069
Seasonal Accident			
Yes	0.693	0.2246	0.1027
No	.	.	.

Note: · p < 0.1. * p < 0.05. ** p < 0.01.

The estimation results provide several insights into the determinants of farmers' seed retention decisions.

Yield per Hectare: The yield per hectare shows a negative association with seed retention. Holding other factors constant, an additional ton per hectare increase in yield reduces the odds of seed retention by approximately 61%. Meaning that farmers with higher yields are likely to prioritize maximizing output and profitability, which may encourage them to rely on purchased or certified seeds rather than reusing part of their harvest as seeds in the next season.

Seed Cost Per Hectare: The seed cost per hectare also shows a negative association with seed retention. An increase of one hundred EGP per hectare in seed cost reduces the odds of seed retention by approximately 72.8%, holding other factors constant. This reflects a clear behavioral pattern: farmers who spend more on seeds are less likely to retain part of their harvest for planting.

The occurrence of farm accidents and machinery cost per hectare were not found to have a significant effect on seed retention decisions.

Overall, the results suggest that seed retention behavior among Egyptian wheat farmers is primarily driven by economic considerations related to productivity and input investment rather than by farm shocks or mechanization costs. This highlights a shift toward more market-oriented production strategies, particularly among higher-performing farmers.

While the logistic regression model provides valuable insights into specific determinants of farmer decision-making, it focuses on a limited set of variables and linear relationships. However, farm-level behavior is often shaped by complex interactions between multiple quantitative and qualitative factors. Therefore, to better capture the multidimensional structure of agricultural practices and farmer characteristics, the analysis proceeds by applying a dimensionality reduction technique that can jointly handle mixed data types.

3.2.5 Factor Analysis of Mixed Data:

Rather than acting as a generic data-reduction tool, FAMD serves as a critical methodological bridge designed to unify the continuous and categorical insights from this study's separate econometric models. Because investigating farm-level wheat dynamics involves a complex mixture of quantitative metrics (such as yields and costs) and qualitative attributes (such as management decisions and institutional support), analyzing these elements in isolation risks overlooking their deeply intertwined reality. Traditional techniques force a fragmentation of the dataset, as PCA handles only continuous variables while MCA is restricted to nominal data. FAMD is essential to the integrity of this research because it integrates both data types into a single matrix, scaling them dynamically to prevent structural bias. Ultimately, employing FAMD is vital to synthesize these distinct analytical pieces into comprehensive, latent dimensions, providing a unified framework to uncover the multi-dimensional farming systems and inform targeted agricultural policy recommendations.

3.2.5.1 Data Preprocessing and Strategic Variable Selection

Before performing the FAMD, the dataset was refined to ensure the model focused on meaningful signal rather than statistical noise. This preprocessing stage was essential for producing interpretable dimensions and reliable visual representations of both variables and individuals.

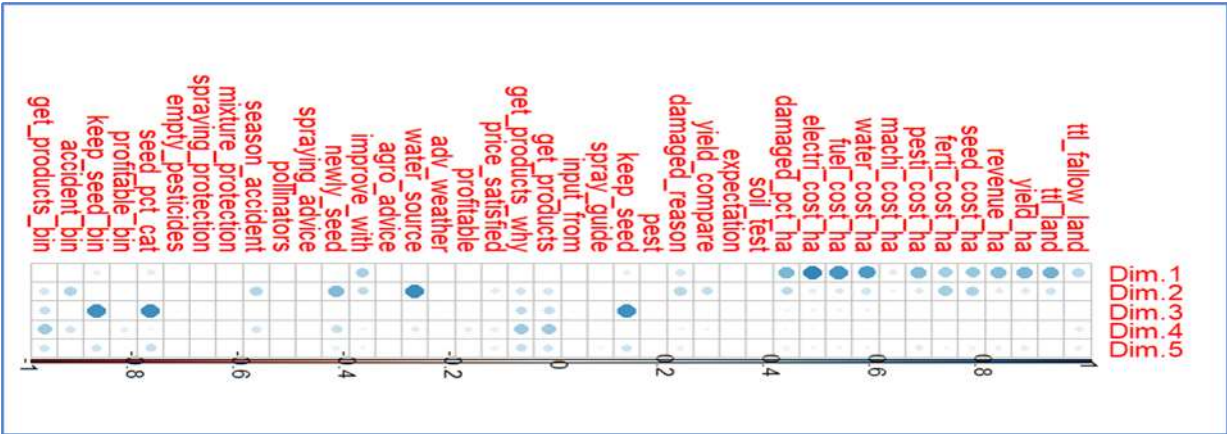


Figure 3.9: Correlation Matrix of Variables and Principal Dimensions.

The initial step involved examining the correlation between all variables and the extracted dimensions using a color-coded correlation matrix. In this matrix, the size and intensity of the circles indicate the strength of the relationship between a variable and a given dimension.

Variables that showed consistently weak correlations across all dimensions were excluded from the final model, as their inclusion would reduce the interpretive clarity of the resulting factor space without contributing meaningful structure. As seen in Figure 3.9, this visual diagnostic allowed for the identification of primary drivers such as land area and per-hectare costs in Dimension 1 while filtering out noise-heavy variables, and categorical levels were "lumped" to prevent rare observations from creating artificial, non-representative dimensions. This process ensured that the final factor space was built on the most representative patterns of the Egyptian wheat sector.

3.2.5.2 Testing the Assumptions of the FAMD and Global Model Fit

The key FAMD assumptions were systematically examined prior to estimation.

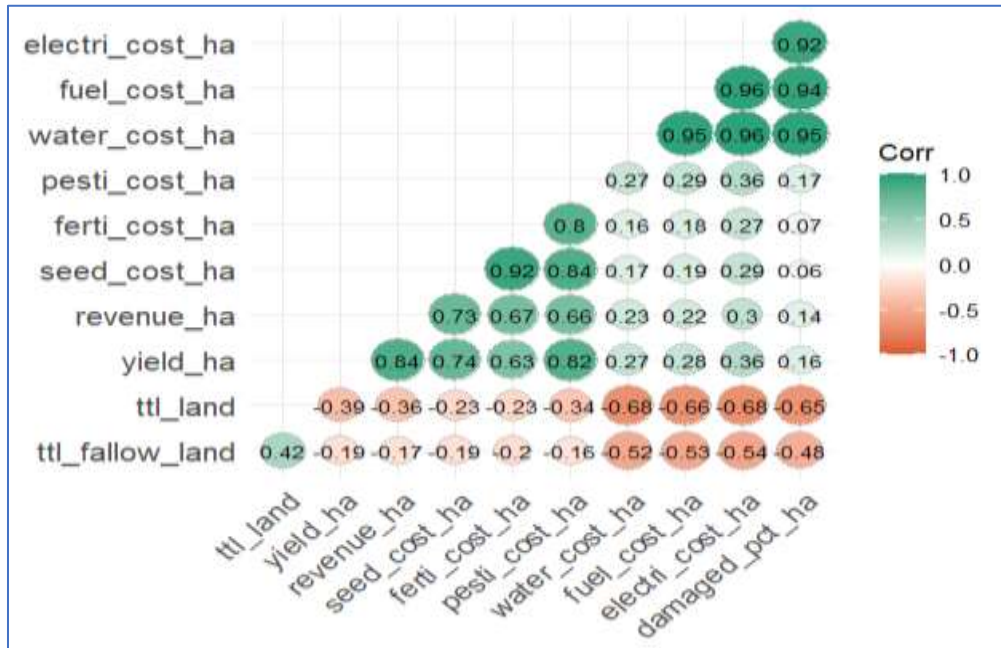


Figure 3.10: Correlation Matrix of Quantitative Variables

- Correlation:** The correlation matrix in Figure 3.10 revealed strong positive associations among quantitative variables, particularly among utility cost variables, where water cost, fuel cost, and electric cost exhibited correlations exceeding 0.94, and between yield per hectare and revenue ($r = 0.84$), confirming sufficient shared variance for factor extraction.
- Linearity:** The assumption was partially met; while some variable pairs displayed approximately linear trends, several relationships exhibited right-skewed patterns, acknowledged as a limitation that is not expected to substantially distort the extracted latent structure given FAMD's distance-based framework.
- Multicollinearity:** VIF values for several cost variables substantially exceeded the threshold of 10. However, as *Tabachnick and Fidell (2013)* established, factor analysis is specifically designed to summarize correlated variables into shared latent dimensions, and multicollinearity is therefore an inherent feature of the data structure being modeled rather than a methodological violation.

The model's performance was evaluated via a Scree Plot, illustrating the percentage of variance explained by each latent dimension.

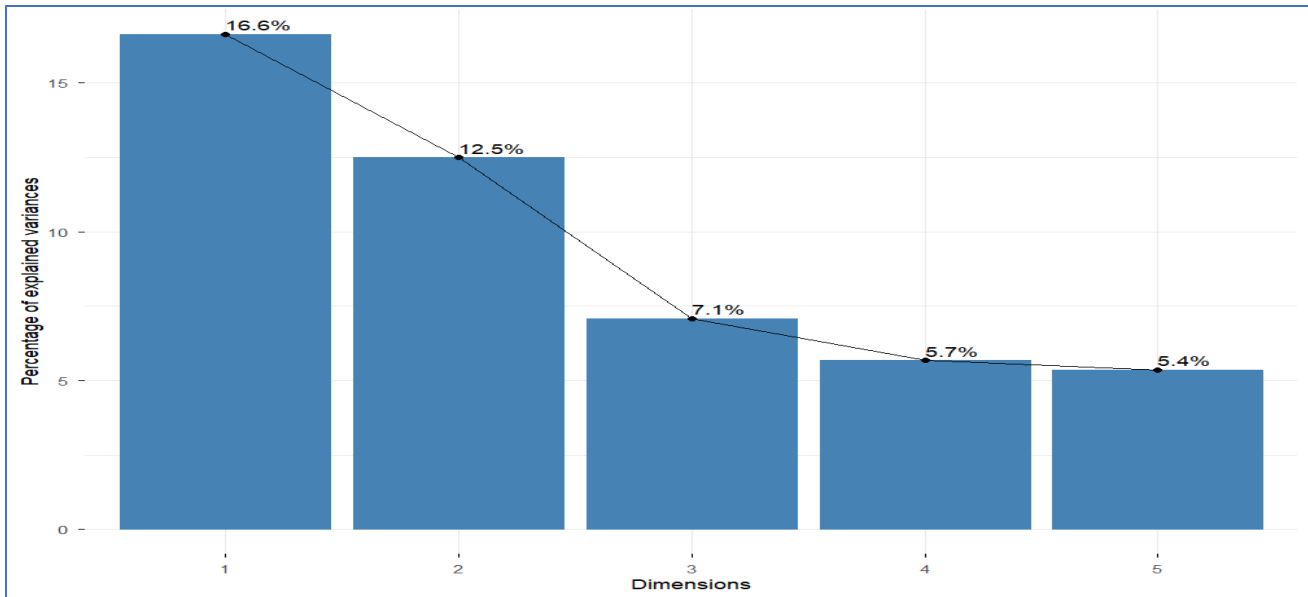


Figure 3.11: Scree Plot of Percentage of Explained Variance.

As shown in Figure 3.11, the first three dimensions collectively capture 36.2% of the total variance. The Scree Plot reveals that the curve begins to flatten after Dimension 2, suggesting that a two-dimension solution could be statistically sufficient according to the elbow criterion. However, Dimension 3 was also retained because it still explains a meaningful proportion of variance (7.1%) and contributes additional interpretive value to the analysis. This supports the retention of three dimensions consistent with best practice guidelines, whereby the number of points above the elbow in the scree plot provides a good estimate of the ideal number of factors to retain *Costello&Osborne, (2005)*. Although 36.2% may appear modest, this is consistent with expectations for mixed datasets spanning multiple behavioral and agronomic domains, where high individual-level heterogeneity naturally limits the concentration of variance in any single dimension. Factors in FAMD should be validated using multiple criteria, including cumulative explained variance, scree plots for eigenvalue drop off, and correlations between factors and external criteria relevant to the domain *Pajès, (2004)*. Accordingly, the validity of the extracted dimensions is judged primarily by their theoretical interpretability rather than by variance thresholds alone, a position consistent with *Watkins (2018)*, who notes that factor solutions in complex research are evaluated based on substantive meaningfulness.

Table 3.13: Eigenvalues of FAMD

Dimension	Eigenvalue
1	5.991
2	4.497
3	2.548
4	2.042
5	1.931

From table 3.13, five dimensions satisfy the Kaiser-Guttman criterion (eigenvalue > 1). This criterion tends to overestimate the number of factors to retain in datasets with many variables, so based on the three ways of retention, collectively only the first three dimensions were retained for interpretation.

3.2.5.3 Results and Interpretation of the Latent Dimensions:

Based on the results of the FAMD, the following sections interpret the three primary dimensions that define the structural characteristics of the wheat sector.

Table 3.14 Variable Coordinates Across the First Three FAMD Dimensions

Variable	Dim.1	Dim.2	Dim.3
Total fallow land	0.288	0.000	0.053
Total land	0.437	0.201	0.020
Yield per Hectare	0.460	0.126	0.007
Total revenue	0.413	0.127	0.003
Seed Cost per Hectare	0.397	0.298	0.000
Ferti. Cost/ha	0.351	0.334	0.000
Pesti. Cost/ha	0.442	0.133	0.001
Water. Cost/ha	0.552	0.205	0.090
Fuel. Cost/ha	0.570	0.177	0.089
Electric. Cost/ha	0.655	0.123	0.090
damage per hectare	0.428	0.287	0.104
Yield compare	0.002	0.246	0.007
Damaged reason	0.192	0.253	0.017
Keep seed	0.117	0.002	0.654
Get products	0.000	0.184	0.134
Get products why	0.004	0.192	0.150
Price satisfied	0.023	0.150	0.003
Profitable	0.020	0.001	0.036
Water source	0.000	0.631	0.002
Improve with	0.352	0.217	0.023
Newly seed	0.034	0.452	0.011
Seasonal accident	0.009	0.312	0.018
Seed pct	0.121	0.004	0.663

To define the relationship between variables and each dimension, a cutoff point of 0.3 was used. This threshold was chosen as it is well-suited to the nature of agricultural data, where diverse environmental and human factors often lead to more distributed variation. Using this limit ensures that all meaningful practical relationships are captured in the analysis.

Dimension 1: Scale and Production Intensity

As shown in Table 3.14, the first dimension is mostly linked to the costs of energy and resources, specifically electricity, fuel, and water. It also includes key factors like crop yield, farm size, pesticide spending, and overall revenue. This dimension separates farmers based on how large and intensive their operations are.

Dimension 2: Farming Strategy and Adaptation

The second dimension is mainly shaped by the source of water and whether farmers buy new seeds. It is also defined by fertilizer costs and environmental challenges, such as seasonal accidents or crop damage. This dimension groups farmers according to how they manage their water and adapt their farming methods. The link between water management and seed choice suggests that this dimension captures a farmer's ability to handle risks. It shows that environmental pressures directly influence how farmers decide to spend money on supplies.

Dimension 3: Seed Saving and Resource Management

The third dimension is almost entirely focused on seed-saving behavior and the amount of seed kept for the next season. Other economic and production factors have very little influence here. This indicates that this dimension represents a strategy of self-reliance and resource conservation. Farmers who score high in this area prefer to save their own seeds rather than buying them every year. This approach helps them reduce their dependence on outside markets and keeps their farming cycle continuously using their own resources.

The three extracted dimensions from the FAMD provide a multidimensional summary of the farming system characteristics previously examined through the econometric models. The first dimension mainly captures the intensity of agricultural input use and production-oriented management practices, which

aligns closely with the Quantile Regression results where pesticide cost, seed investment, irrigation source, and land utilization emerged as important determinants of wheat yield, particularly among high-performing farmers. The second dimension reflects post-harvest management behavior, crop damage, and safety-related practices, supporting the earlier findings regarding the role of post-harvest conditions and damaged crop percentages in shaping farm outcomes and productivity variability. The third dimension appears to represent economic and behavioral decision-making patterns among farmers, particularly profitability status and seed retention behavior, which corresponds with the Binary Logistic Regression models used to explain profitable operations and farmer seed-saving decisions.

Overall, the FAMD results complement the regression analyses by revealing that wheat farming outcomes in Egypt are not driven by isolated variables alone, but rather by interconnected latent structures combining agronomic practices, post-harvest management, and economic behavior. This confirms the multidimensional nature of wheat production systems and supports the need for integrated agricultural policies rather than one-dimensional interventions.

Chapter (4)

Conclusions and Recommendations

4.1 Conclusions:

When we started this study, the question seemed straightforward: what makes some Egyptian wheat farmers more productive than others? But the deeper we looked, the clearer it became that there is no simple answer. Egyptian wheat farming is not one story. It is hundreds of different stories, shaped by water access, financial pressure, seasonal luck, and decisions made under uncertainty every single season.

What we found is that the farmers who struggle are not struggling because they are doing something wrong. They are struggling because the systems around them, irrigation infrastructure, input markets, and advisory programs were often not designed with their specific realities in mind. A spraying guideline written for ideal conditions does not help a farmer dealing with poor soil and unreliable water. A certified seed program helps a farmer who can afford it, not one who is already stretched thin.

At the same time, the farmers who are performing well face a different kind of vulnerability. One bad season, one pest outbreak, one accident, and years of hard work can be wiped out. Post-harvest losses are not just a statistic. They represent real income, real food, and real livelihoods lost after all the hard work of growing the crop has already been done.

The clearest finding of this study is also the simplest one: pesticide investment pays off for every type of farmer. That is rare. Most things in agriculture depend heavily on context. But affordable crop protection is as close to a universal recommendation as this data allows us to make.

Beyond that, the message is less about what to do and more about how to think. Egyptian wheat farmers deserve policies that see them as individuals with different needs, not as a single group with a single problem. A farmer at the bottom of the productivity distribution needs a completely different kind of support than a farmer at the top and pretending otherwise is not just ineffective; it can actively make things worse.

Egypt is one of the world's largest wheat importers. That is not inevitable. The potential to grow more, lose less, and support farmers better is there. This study is a small contribution toward understanding how.

4.2 Policy Recommendation:

Invest in Irrigation Infrastructure Rain-fed farming proved advantageous only for low-yield farmers, while becoming a constraint for higher performers. Simultaneously, reliance on rain-fed agriculture was associated with greater post-harvest loss risk. Improving access to reliable, efficient, and cost-effective irrigation systems should therefore be treated as a foundational priority, one that benefits farmers across all productivity levels and reduces exposure to environmental variability.

Improve Affordability and Accessibility of Crop Protection Inputs Pesticide investment emerged as the single most consistent and universal driver of wheat yield across all farmer types. Any policy that improves the affordability and accessibility of crop protection inputs will generate positive returns regardless of where a farmer sits in the productivity distribution.

Adopt Differentiated and Locally Adapted Extension Services A uniform advisory approach is insufficient given the heterogeneous nature of Egyptian wheat farming. Extension services should be tailored to each farmer's productivity level, resource availability, and local conditions. For low-yield farmers, externally designed buyer guidelines were found to negatively affect productivity, highlighting the urgent need for locally adapted programs that reflect the realities of smallholder farming rather than imposing generic protocols.

Prioritize Fundamental Constraints before Advanced Interventions For low-yield farmers, investments in water access and affordable input must come first. Seed quality improvements, advanced agronomic practices, and land optimization strategies can only generate meaningful returns once these foundational constraints are addressed. Policymakers should therefore sequence their interventions carefully, ensuring that struggling farmers receive support for their most pressing needs before more sophisticated programs are introduced.

Reduce Post-Harvest Losses Crop damage showed a direct and significant negative effect among farmers, while seasonal accidents were found to increase post-harvest loss risk across the sample. Targeted interventions, including improved storage facilities, better harvesting techniques, post-harvest management training, crop insurance programs, and early warning systems for seasonal disruptions, are essential to protect productivity gains already achieved in the field.

Facilitate Access to Quality Seeds and Reduce Retention Dependency Farmers already exhibit market-oriented seed purchasing behavior, but financially constrained farmers often resort to seed retention as a cost-saving strategy. Policies should simultaneously subsidize certified seed access for smallholder farmers, develop locally accessible and competitively priced seed markets, and promote awareness of the agronomic risks associated with repeated seed retention such as genetic degradation and disease transmission to gradually shift all farmers toward improved seed use.

Strengthen Risk Management and Farm Resilience Seasonal accidents represent a significant source of economic disruption for Egyptian wheat farmers. Developing comprehensive risk management frameworks, including crop insurance programs, climate-resilient farming practices, and adaptive field management strategies, would reduce the economic impact of seasonal shocks and improve overall farm stability.

Promote Efficient Input Use and Cost Optimization Higher total production costs were associated with greater profitability when inputs were used effectively, suggesting that the key is not spending less but spending wisely. Agricultural programs should therefore focus on helping farmers optimize their input use, particularly fertilizers, seeds, machinery, and pest control, to maximize returns on investment rather than simply increasing expenditure.

Enhance Integrated Pest Management Farms affected by rodents and pest pressure showed lower profitability and higher loss risk. Agricultural extension programs should emphasize integrated pest management strategies, providing farmers with practical training on effective pest and rodent control methods to reduce crop losses both at the field and post-harvest stages.

References:

1. Abdalla, A., Becker, M., & Stellmacher, T. (2023). The contribution of agronomic management to sustainably intensify Egypt's wheat production. *Agriculture*, 13(5), 978. <https://doi.org/10.3390/agriculture13050978>, [Accessed: 2 Dec 2025]
2. Agresti, A. (2013). *Categorical Data Analysis* (3rd ed.). John Wiley & Sons. <https://mybiostats.wordpress.com/wp-content/uploads/2015/03/3rd-ed-alan-agresti-categorical-data-analysis.pdf>, [Accessed: 20 Feb 2025]
3. Alboghdady, M., Abbas, S., Alashry, M. K., Hu, Y., & El-Hendawy, S. (2025). Modeling the short- and long-term impacts of climate change on wheat production in Egypt using autoregressive distributed lag approach. *Land*, 14(10), 1962. <https://doi.org/10.3390/land14101962>, [Accessed: 3 Dec 2025]
4. Belsley, D. A., Kuh, E., & Welsch, R. E. (1980). *Regression diagnostics: Identifying influential data and sources of collinearity*. New York: John Wiley & Sons.
5. Bucheli, J., Visse-Mansiaux, M., Herrera, J., Levy Häner, L., Tack, J., & Finger, R. (2024). Precipitation causes quality losses of large economic relevance in wheat production. *Q Open*, 4(1), qoae008. <https://doi.org/10.1093/qopen/qoae008>, [Accessed: 5 Dec 2025]
6. Burnham, K. P., & Anderson, D. R. (2002). *Model selection and multimodel inference: A practical information-theoretic approach* (2nd ed.). Springer.
7. Byamungu et al. (2026) — "Seed Choices and Economic Outcomes: Evidence from Wheat Farmers in Tanzania" — published in *Cogent Food & Agriculture* (Taylor & Francis). <https://www.tandfonline.com/doi/full/10.1080/23311932.2026.2668771?af=R#abstract>, [Accessed: 8 Dec 2025]
8. Cattell, R. B. (1966). The scree test for the number of factors. *Multivariate Behavioral Research*, 1(2), 245–276. <https://garfield.library.upenn.edu/classics1983/A1983PY50000001.pdf>, [Accessed: 2 Mar 2026]
9. Costello, A. B., & Osborne, J. W. (2005). Best practices in exploratory factor analysis: Four recommendations for getting the most from your analysis. *Practical Assessment, Research, and Evaluation*, 10(7), 1–9. <https://www.researchgate.net/publication/209835856>, [Accessed: 1 Mar 2026]
10. El Bahgy, E. E. K. M., Bahloul, A. M. E., El Khishin, E. A. E., Yigezu, A. Y., & Gado, E. H. M. (2024). Evaluating the economic impact of agricultural technology adoption on wheat yield improvement. *Scientific Journal of Agricultural Sciences*, 6(4), 213–225. https://sjas.journals.ekb.eg/article_417932.html, [Accessed: 2 Dec 2025]
11. El-Batran, E. S., Attia, S. M. R. I., & Alarag, S. A. E. (2023). An economic study of wheat postharvest losses in Ismailia Governorate. *Alexandria Science Exchange Journal*, 44(1), 113–133. https://asejaiqsae.journals.ekb.eg/article_292819.html, [Accessed: 3 Dec 2025]
12. El Wakeel, S. M. A. H., & Yassa, N. S. Y. (2025). The impact of wheat cultivation methods on total factor productivity in Beheira Governorate. *Alexandria Science Exchange Journal*, 46(3), 483–494. <https://doi.org/10.21608/asejaiqsae.2025.450872>, [Accessed: 15 Dec 2025]
13. Escofier, B., & Pagès, J. (1994). Multiple factor analysis (AFMULT package). *Computational Statistics & Data Analysis*, 18(1), 121–140. <https://three-mode.leidenuniv.nl/pdf/e/escofierpages1994csda.pdf>, [Accessed: 4 Mar 2026]
14. Fawcett, T. (2006). An introduction to ROC analysis. *Pattern Recognition Letters*, 27(8), 861–874. <https://people.inf.elte.hu/kiss/13dwhdm/roc.pdf>, [Accessed: 22 Feb 2025]
15. Food and Agriculture Organization of the United Nations (FAO). (2019). *The State of Food and Agriculture 2019: Moving forward on food loss and waste reduction*. FAO. <https://www.fao.org/interactive/state-of-food-agriculture/2019/en/>, [Accessed: 29 Nov 2025]

16. Food and Agriculture Organization of the United Nations (FAO). (2022–2023). *Good Growth Plan microdata for Egypt*.
https://microdata.fao.org/index.php/catalog/2623/data-dictionary/F33?file_name=EGY_PostHarvest-Wheat. [Accessed: 25 Oct 2025]
17. Green, S. B. (1991). How many subjects does it take to do a regression analysis? *Multivariate Behavioral Research*.
<https://www.scribd.com/document/504497572/Green-1991>, [Accessed: 2 Feb 2026]
18. Greene, W. H. (2003). *Econometric Analysis* (5th ed.). Prentice Hall.
https://www.ctanujit.org/uploads/2/5/3/9/25393293/_econometric_analysis_by_greence.pdf, [Accessed: 1 Feb 2026]
19. Huang, T., Li, B., Shen, D., Cao, J., & Mao, B. (2017). Analysis of the grain loss in harvest based on logistic regression. *Procedia Computer Science*, 124, 97–103.
<https://www.sciencedirect.com/science/article/pii/S1877050917326741>, [Accessed: 16 Dec 2025]
20. Hussain, B., Ali, U., Shaheen, S., & Almas, L. K. (2023). Climate volatility, wheat productivity and food security: A quantile regression analysis. *WSEAS Transactions on Environment and Development*, 19, 1196–1209. <https://doi.org/10.37394/232015.2023.19.109>, [Accessed: 2 Feb 2026]
21. Hosmer, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). *Applied Logistic Regression* (3rd ed.). John Wiley & Sons. <https://pzs.dstu.dp.ua/DataMining/mls/bibl/logistic.pdf>, [Accessed: 20 Feb 2025]
22. Koenker, R. (2005). *Quantile Regression*. Cambridge University Press.
https://box.cs.istu.ru/public/docs/other/_Unsorted/new/books.pdcox.net/Math/Quantile%20Regression.pdf, [Accessed: 5 Feb 2026]
23. McCullagh, P., & Nelder, J. A. (1989). *Generalized Linear Models* (2nd ed.). Chapman & Hall.
<https://utstat.utoronto.ca/~brunner/oldclass/2201s11/readings/glmbook.pdf>, [Accessed: 4 Feb 2026]
24. Mieke, C., Nabwire, L., Sparrow, R., Spielman, D., & Van Campenhout, B. (2023). ‘Miracle seeds,’ informational curses? The risk of high expectations for new agricultural technologies. *IFPRI Blog*.
<https://www.ifpri.org/blog/miracle-seeds-informational-curses-risk-high-expectations-new-agricultural-technologies/>, [Accessed: 2 Dec 2025]
25. Minwagaw (2021) — “Determinants of Seed Distribution System: The Case of Womberma District, Northwest Ethiopia” — published in *Advances in Agriculture* (Wiley).
<https://onlinelibrary.wiley.com/doi/10.1155/2021/3656320>, [Accessed: 9 Dec 2025]
26. Nath, P., Singh, R., & Kumar, A. (2024). *Research and technologies to reduce grain postharvest losses: A review*. Retrieved from
<https://www.researchgate.net/publication/381456218>, [Accessed: 5 Dec 2025]
27. Pagès, J. (2004). Analyse factorielle de données mixtes. *Revue de Statistique Appliquée*, 52(4), 93–111.
https://www.numdam.org/article/RSA_2004__52_4_93_0.pdf, [Accessed: 3 Mar 2026]
28. Pagès, J. (2014). *Multiple Factor Analysis by Example Using R*. Chapman and Hall/CRC.
https://api.pageplace.de/preview/DT0400.9781482205480_A24097362/preview-9781482205480_A24097362.pdf, [Accessed: 3 Mar 2026]
29. Shaheen, M. M. E. S., Salem, M. H., AbdelMaqsood, A. M., & AbdelFatah, I. A. M. (2025). Economic and marketing efficiency of wheat crop in Egypt. *Egyptian Science Magazine*, 12(1), 87–96.
<https://doi.org/10.21608/esm.2025.452240>, [Accessed: 5 Dec 2025]
30. Shao, Q., & Wang, Y. G. (2009). Statistical power calculation and sample size determination for environmental studies with data below detection limits. *Water Resources Research*, 45(9), W09410.
<https://www.researchgate.net/publication/232256857>, [Accessed: 3 Feb 2026]
31. Vishwakarma, S., Zhang, X., & Lyubchich, V. (2025). Unveiling the drivers contributing to global wheat yield shocks through quantile regression. *Artificial Intelligence in Agriculture*, 15(3), 564–572.
<https://doi.org/10.1016/j.aiaa.2025.03.004>, [Accessed: 10 Dec 2025]

32. **Watkins, M. W. (2018).** Exploratory factor analysis: A guide to best practice. *Journal of Black Psychology*, 44(3), 219–246.
https://journals.sagepub.com/doi/full/10.1177/0095798418771807?_gl=1*1n6lqy6*_up*MQ..*_ga*NjczMDc0NDc2LjE3Nzg4NDM1MjE.*_ga_60R758KFDG*czE3Nzg4NDM1MjAkczEkZzEkdDE3Nzg4NDM1NjgkajEyJGwwJGg3ODcyNTg2MTY ,[Accessed: 5 Mar 2026]
33. **World Bank. (2023).** *The nexus of food, water & energy: A key element to Egypt's climate efforts.*
<https://www.worldbank.org/en/news/feature/2023/12/18/the-nexus-of-food-water-energy-a-key-element-to-egypt-s-climate-efforts> ,[Accessed: 16 Dec 2025]
34. **Yigezu, Y. A., Moustafa, M. A., Mohiy, M. M., Ibrahim, S. E. D., Ghanem, W. M., Niane, A. A., Abbas, E., Sabry, S. R. S., & Halila, H. (2021).** Food losses and wastage along the wheat value chain in Egypt and their implications on food and energy security, natural resources, and the environment. *Sustainability*, 13(18), 10011.
<https://www.mdpi.com/2071-1050/13/18/10011> , [Accessed: 5 Dec 2025]